A Comparative Study of Physics-Augmented and Projection-Based Surrogates for Anisotropic Hyperelastic RVE Homogenization

S. Ares De Parga[1,2] A. Cornejo[1,2] J. A. Hernández[1,2] R. Rossi[1,2]

[1]Department of Civil and Environmental Engineering, Universitat Politècnica de Catalunya (UPC), Barcelona, Spain

[2]Centre Internacional de Mètodes Numèrics en Enginyeria (CIMNE), Barcelona, Spain

July 30, 2026

Abstract

Physics-augmented neural networks (PANNs) for hyperelastic constitutive modelling are commonly held to a six-point admissibility checklist: (1) thermodynamic consistency, (2) objectivity, (3) tensor symmetry, (4) normalization at the reference configuration, (5) polyconvexity, and (6) a growth/coercivity condition [4, 1]. This paper works through each point in full mathematical detail for a single certified model of the two-dimensional nonlinear hyperelastic representative volume element (RVE) treated in this project, an energy-based PANN built from an input-convex neural network (ICNN) acting on balanced directional invariants of F and $\operatorname{cof} F$. Five of the six points are satisfied exactly, by construction, with a closed-form proof given below for each. The sixth, polyconvexity in the sense of Ball [1], is proved by an explicit construction rather than a numerical penalty; we also verify numerically the weaker rank-one/Legendre–Hadamard condition it implies, and separately show that a strictly stronger, globally SPD-tangent condition – used elsewhere in the literature (e.g. As'ad, Avery and Farhat [5]) for a different anisotropic RVE, for its own well-documented reasons – is not implied by polyconvexity and is, for this RVE's own full-order model (FOM), directly contradicted by the measured ground-truth tangent. The model is trained on Stage–1 trajectories only, evaluated once on the untouched Stage–10 trajectory, and compared against a projection-based hyper-reduced-order model (HPROM–ANN, in both its iterative-corrector and direct-evaluation operating modes) on the same path, and against a direct strain-to-stress regression baseline with no energy potential at all, which is shown to violate thermodynamic consistency directly (a nonzero cyclic-work integral around closed strain loops) despite matching or exceeding every certified model's held-out accuracy. A state-of-the-art survey of seven directly relevant physics-augmented neural network (PANN) constructions is given, precisely comparing which admissibility properties each achieves exactly, approximately, or not at all; it identifies, in the surveyed authors' own words, an explicit open problem – combining exact objectivity, an exact stress-free reference state, and exact polyconvexity for an anisotropic material with no assumed symmetry group – that the construction here resolves for this RVE. The same proofs are shown to transfer verbatim to a second convex core, an input-convex Kolmogorov–Arnold network, at comparable accuracy with far fewer parameters; and a matched, otherwise identical, uncertified flexible PANN is used to show directly that held-out accuracy cannot detect the absence of a polyconvexity certificate – it matches or beats the certified models on every accuracy metric while failing a rank-one convexity check on 7% of sampled, ordinary deformation states.

Contents

1 Introduction 2

2 Background: full-order, projection-based, and hyper-reduced models 3

2.1 The representative volume element and its full-order model 4

2.2 Projection-based reduced-order model (PROM) 5

2.3 Nonlinear-manifold correction: the HPROM–ANN decoder 5

2.4 Hyperreduction: the empirical cubature method and MAW–ECM 6

3 The admissibility checklist 7

4 Related work: what is proven, what is imprecise, and what remains open 8

4.1 Models that enforce a strictly stronger condition than polyconvexity 8

4.2 Models that enforce Ball's polyconvexity, but only for a named symmetry group 9

4.3 A model that widens the anisotropy class by giving up polyconvexity . . . 10
4.4 Isotropic and fiber-restricted polyconvex CANNs, and the growth condition in practice . . . 10
4.5 Where published claims have been documented to be wrong . . . 10
4.6 What remains open, and what this work contributes . . . 11

5 Kinematics and stress convention 11

6 (C1) Thermodynamic consistency 11

7 A pure-regression baseline: violating (C1) directly 12

8 (C2) Objectivity 13

9 (C3) Symmetry: two different meanings 13

10 (C4) Normalization: the stress-free reference state 14
10.1 The energy value: trivial by construction . . . 14
10.2 The stress value: not trivial, and this is where the balanced directions matter . . . 14

11 (C5) Polyconvexity 15
11.1 Definition . . . 15
11.2 A strictly stronger alternative used elsewhere, and why it is not adopted here . . . 15
11.3 The necessary condition we verify: rank-one convexity . . . 16
11.4 The construction, and the proof that it is polyconvex . . . 16

12 (C6) Growth condition 17

13 Loss design: an equivalent-stress-oriented training choice 18

14 Training protocol 18

15 Held-out Stage–10 assessment: all six model variants 19

16 A second convex core: input-convex Kolmogorov–Arnold network (ICKAN) 20

17 What breaks without the certificate: a non-polyconvex baseline 21

18 Conclusions 22

1 Introduction

Multiscale (FE^2-type) computational homogenization resolves each macroscopic material point by solving a nonlinear boundary value problem on a representative volume element (RVE), so that the macroscopic constitutive response is not postulated but computed from the microstructure directly. This is expensive: every macroscopic Gauss point, at every load step and every Newton iteration, requires its own converged microscale solve. Two largely separate lines of work address this cost. The first replaces the microscale solve by a data-driven, physics-augmented neural network (PANN) surrogate for the homogenized constitutive law itself, trained once offline on precomputed RVE solves. The second instead reduces the dimensionality of the microscale problem directly, via projection-based (PROM) and hyper-reduced-order (HPROM) models that keep the RVE's finite element structure but solve it in a low-dimensional subspace with a sparse, hyperreduced integration rule. Both routes are active research areas in their own right, and both are used in this paper – not as competitors, but as two independent, non-interchangeable sources of ground truth and comparison for the same RVE, evaluated on the same held-out load path.

For the first route, a specific and recurring difficulty motivates this work. Physics-augmented neural constitutive models are commonly held to an admissibility checklist – (C1) thermodynamic consistency, (C2) objectivity, (C3) tensor symmetry, (C4) a stress-free reference state, (C5) polyconvexity, and (C6) a growth condition – and the current literature (surveyed in full in Section 4) shows a clear, recurring trade-off: models that enforce every one of these properties *exactly*, by construction, do so only for a material with a *named* symmetry class (isotropic, transversely isotropic, cubic, . . .); models that relax the symmetry assumption to a genuinely general anisotropic material give up polyconvexity, the reference state, or exact objectivity to do so. Linden et al. state explicitly that a complete, polyconvexity-compatible invariant set may not exist for some symmetry groups, and flag a general, symmetry-agnostic construction as an open problem. Whether this trade-off is fundamental, or an artifact of how the anisotropy has been encoded so far, is the central question this paper answers for a concrete, fully worked 2D RVE: a Neo-Hookean solid with a hole, admitting no assumed material symmetry.

Answering that question carefully also requires knowing, precisely, what "accurate" does and does not certify. A model can match or exceed a certified model's held-out accuracy while silently violating thermodynamic consistency or losing rank-one convexity at ordinary deformation states – this is not a hypothetical concern but something demonstrated directly on this project's own data in Sections 7 and 17. This is also why the paper evaluates six model variants side by side on the identical held-out Stage–10 path, spanning a genuine spectrum of embedded physical structure: a pure regression surrogate with no potential at all, a free (uncertified) energy surrogate, two independently certified polyconvex cores (ICNN and ICKAN), and the two operating modes of HPROM–ANN, a projection-based model that satisfies every admissibility property automatically because it solves (a hyperreduced form of) the true microscale equilibrium equations rather than learning a constitutive law at all.

Contributions. This paper's own contributions are specifically:

1. A closed-form construction of a PANN for a general, group-free anisotropic hyperelastic material that satisfies (C1)–(C6) exactly, by construction, including exact polyconvexity in Ball's sense (Section 11.4) – to our knowledge the first construction to combine exact objectivity, an exact stress-free reference state, and exact polyconvexity without assuming or exploiting any named material symmetry class (Section 4.6). We also verify the correct necessary consequence of polyconvexity (rank-one convexity) numerically, and show that a strictly stronger, globally SPD-tangent condition used elsewhere in the literature for a different material is not required by polyconvexity and is, for this RVE's own ground truth, false (Section 11.2).

2. A direct, quantitative demonstration that held-out accuracy alone cannot certify admissibility: a pure-regression baseline with no energy potential is the *most* accurate of all six model variants while directly violating (C1) (Section 7), and an otherwise-identical uncertified free model matches the certified models' accuracy while failing a rank-one convexity check on 7% of sampled states (Section 17).

3. A reaction-force-based procedure for recovering the exact, energy-conjugate homogenized stress from a converged HPROM–ANN state (Section 15), needed because this quantity is not, in general, the naive volume average of the microscopic stress field for a mixed Dirichlet/free boundary-value problem.

4. Verification that the same certificate transfers, proof-for-proof, to a second convex core (an input-convex Kolmogorov–Arnold network, Section 16) at comparable accuracy with substantially fewer parameters.

The full-order model, the linear and nonlinear projection-based reduced models, and the empirical-cubature hyperreduction used to build HPROM–ANN (Section 2) are established tools from the literature, applied here but not contributed by this work; they are cited to their sources throughout, and Section 15 states this provenance explicitly.

Outline. Section 2 fixes notation for the FOM, PROM, and HPROM–ANN used as an independent, non-ANN-constitutive-law comparison point throughout. Section 3 states the admissibility checklist precisely. Section 4 surveys seven directly comparable PANN constructions against every checklist item, establishing exactly what is already proven, what is imprecise, and what remains open. Section 5 fixes kinematic and stress conventions, and Sections 6–12 take each checklist item in turn, with Section 11.4 giving the explicit polyconvex construction. Section 13 and Section 14 describe the training loss and protocol. Section 15 reports the held-out Stage–10 comparison of all six model variants. Section 16 repeats the polyconvexity proofs for the ICKAN core, and Section 17 uses an uncertified baseline to show directly what the certificate protects against. Section 18 concludes.

2 Background: full-order, projection-based, and hyper-reduced models

This section fixes notation and gives a self-contained, high-level account of the full-order model (FOM), the projection-based reduced-order model (PROM), and the hyper-reduced-order model (HPROM) used throughout this paper as an independent, non-ANN-constitutive-law comparison point for the PANN. None of the constructions in this section are a contribution of this work; they are established tools, cited to their sources, applied here to a specific RVE.

2.1 The representative volume element and its full-order model

The RVE is a square unit cell with a central circular hole, Fig. 1, discretized by a standard displacement-based finite element mesh and governed by a single homogeneous, isotropic, compressible Neo-Hookean solid ($E = 1.628\,\text{GPa}$, $\nu = 0.4$). A macroscopic strain $\boldsymbol{\varepsilon} = (E_{11}, E_{22}, \gamma_{12})$ is imposed through an affine Dirichlet condition on the outer boundary nodes (Fig. 1, red), while the remaining boundary and interior nodes – including the nodes on the hole – are left free. We write $\boldsymbol{u}$ for the full nodal displacement field over every node of the mesh (this is what is plotted in Fig. 2), partitioned into its Dirichlet and free parts as $\boldsymbol{u} = (\boldsymbol{u}_d, \boldsymbol{u}_f)$: on the Dirichlet boundary, $\boldsymbol{u}_d$ is prescribed exactly, node by node, by the affine macro-strain map determined by $\boldsymbol{\varepsilon}$ (linear in the reference-configuration position $\boldsymbol{X}$ of each Dirichlet node and in the three components $E_{11}, E_{22}, \gamma_{12}$ of $\boldsymbol{\varepsilon}$), while $\boldsymbol{u}_f \in \mathbb{R}^{n_L}$, the free-node values, are unknown. The finite element (Galerkin) discretization of the governing balance of linear momentum reduces, in the quasi-static setting used here (no inertia, no explicit time dependence – the "Stage–10" load path is a pseudo-time sequence of equilibrium states, not a genuine initial-value problem), to the nodal equilibrium residual

$$\boldsymbol{r}(\boldsymbol{u}_f;\boldsymbol{\varepsilon}) := \sum_{e=1}^{n_{\text{el}}} \boldsymbol{L}_e^T \left(\sum_{g=1}^{n_{\text{gp}}^{(e)}} \boldsymbol{B}_{e,g}^T\, \boldsymbol{S}_{e,g}(\boldsymbol{u}_f;\boldsymbol{\varepsilon})\, w_{e,g} \right) = \boldsymbol{0}, \tag{1}$$

where $\boldsymbol{L}_e$ is the Boolean element-assembly operator, $\boldsymbol{B}_{e,g}$ is the element strain-displacement operator, $\boldsymbol{S}_{e,g}$ is the second Piola–Kirchhoff stress at Gauss point g of element e (evaluated from the true microscopic Neo-Hookean law, not a surrogate, and depending on $\boldsymbol{u}_f$ and the fixed, prescribed $\boldsymbol{u}_d(\boldsymbol{\varepsilon})$ together), and $w_{e,g}$ is the corresponding FE quadrature weight (parent-domain weight times the isoparametric Jacobian determinant). Equation (1) is solved by Newton–Raphson for $\boldsymbol{u}_f$ at every prescribed $\boldsymbol{\varepsilon}$ along a load path; this is the FOM, and it is what generates every training snapshot and every held-out ground-truth number (energy, stress, and their components) used elsewhere in this paper. Throughout this paper, $\boldsymbol{\varepsilon}$ denotes this prescribed, exact boundary strain: the deformation gradient $\boldsymbol{F} = \boldsymbol{U} = \sqrt{2\boldsymbol{E}(\boldsymbol{\varepsilon}) + \boldsymbol{I}}$ (the unique symmetric, rotation-free square root) is used to build the affine map $\boldsymbol{u}_d(\boldsymbol{X}) = (\boldsymbol{F} - \boldsymbol{I})\boldsymbol{X}$ directly, so the outer boundary's own Green-Lagrange strain equals $\boldsymbol{\varepsilon}$ exactly, by construction – unlike the homogenized macro-*stress*, which is a computed output with more than one admissible definition. The homogenized macro-stress reported throughout is *not* the naive volume average of $\{\boldsymbol{S}_{e,g}\}$ (see the distinction drawn in Section 15), but the reaction-force-based, exactly energy-conjugate quantity described there; a volume-averaged homogenized *strain* of the converged solution can likewise be computed and differs measurably from $\boldsymbol{\varepsilon}$ for this mixed Dirichlet/free RVE (the free hole boundary is not affine-constrained, so the classical boundary-average identity that would equate the two does not apply), but this paper does not use that quantity anywhere: $\boldsymbol{\varepsilon}$, being the independent (prescribed) variable rather than a computed output, requires no such choice.

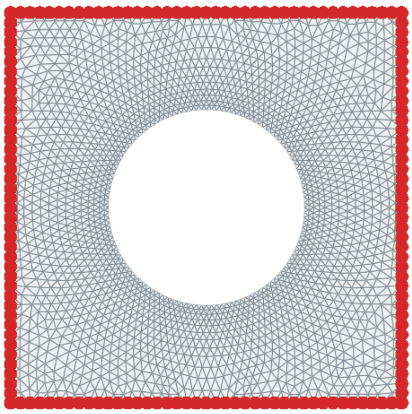

Figure 1: The RVE: a square unit cell with a central circular hole, meshed with standard displacement-based finite elements. The outer boundary (red) carries the prescribed affine macro-strain Dirichlet condition; the remaining boundary and interior nodes, including the hole boundary, are free. The cell geometry has D_4 (square) symmetry, but the PANN construction in this paper does not assume or exploit it – see Section 4.2.

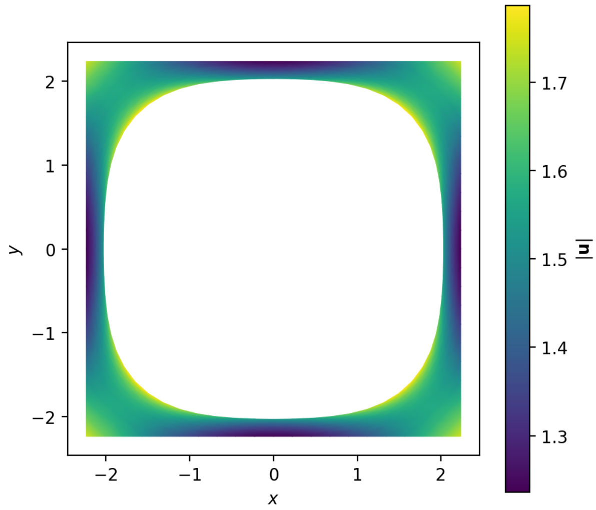

Figure 2: The full displacement field $\boldsymbol{u} = (\boldsymbol{u}_d, \boldsymbol{u}_f)$ reconstructed from a representative converged FOM solution $\boldsymbol{u}_f$ (free part) and the prescribed affine Dirichlet data $\boldsymbol{u}_d$ (boundary part), plotted on the deformed configuration and colored by the nodal displacement magnitude $\|\boldsymbol{u}\|$ (reference coordinates $x, y \in [-1, 1]$; here at the equibiaxial macro-strain state $E_{11} = E_{22} = 2$, $\gamma_{12} = 0$, the last Stage–1 training step shown in Fig. 3 for trajectory 1). The initially circular hole deforms toward a rounded square under equibiaxial stretch, consistent with the D_4 symmetry of Fig. 1.

2.2 Projection-based reduced-order model (PROM)

A standard linear PROM replaces $\boldsymbol{u}_f$ by its projection onto a low-dimensional subspace. Snapshots $\boldsymbol{u}_f(\boldsymbol{\varepsilon}_1), \ldots, \boldsymbol{u}_f(\boldsymbol{\varepsilon}_{n_s})$ from FOM solves along the Stage–0/Stage–1 training trajectories (Fig. 3) are assembled into a snapshot matrix $\boldsymbol{D} \in \mathbb{R}^{n_L \times n_s}$, and a thin, truncated singular value decomposition gives an orthonormal reduced-order basis (ROB) $\mathbf{V}_{\mathrm{tot}} \in \mathbb{R}^{n_L \times n_{\mathrm{tot}}}$ spanning the dominant left singular subspace of $\boldsymbol{D}$. The displacement field is then approximated by

$$\boldsymbol{u}_f \approx \boldsymbol{u}_f^{\mathrm{aff}}(\boldsymbol{\varepsilon}) + \mathbf{V}_{\mathrm{tot}}\, \boldsymbol{q}_{\mathrm{tot}}(\boldsymbol{\varepsilon}), \tag{2}$$

where $\boldsymbol{u}_f^{\mathrm{aff}}(\boldsymbol{\varepsilon})$ is the affine baseline consistent with the prescribed macroscopic Dirichlet data $\boldsymbol{u}_d(\boldsymbol{\varepsilon})$ (so that $\mathbf{V}_{\mathrm{tot}}\boldsymbol{q}_{\mathrm{tot}}$ carries only the correction beyond that affine guess) and $\boldsymbol{q}_{\mathrm{tot}} \in \mathbb{R}^{n_{\mathrm{tot}}}$ are the generalized (POD) coordinates.

Substituting (2) into (1) and testing with the same basis (Galerkin projection – the natural choice here since the tangent operator of a hyperelastic problem is symmetric, unlike the non-self-adjoint operators that motivate a least-squares Petrov–Galerkin, LSPG, test space in the compressible-flow PROM literature [17, 18]) gives the reduced equilibrium equation

$$\mathbf{V}_{\mathrm{tot}}^T\, \boldsymbol{r}\big(\boldsymbol{u}_f^{\mathrm{aff}}(\boldsymbol{\varepsilon}) + \mathbf{V}_{\mathrm{tot}}\boldsymbol{q}_{\mathrm{tot}}; \boldsymbol{\varepsilon}\big) = \mathbf{0}, \tag{3}$$

a system of $n_{\mathrm{tot}} \ll n_L$ nonlinear equations, again solved by Newton–Raphson. As documented for this exact class of problem (propagating features, evolving nonlinear regions) by Barnett and Farhat [16] and by Hernández's own metamaterial and damage examples [19], a single linear subspace of dimension small enough to be worth reducing to is generally *not* enough to reach high accuracy for a nonlinear problem (the Kolmogorov n-width barrier): the number of retained modes needed for good accuracy can remain uncomfortably large. The nonlinear-manifold construction used by HPROM–ANN in this paper addresses exactly this point.

2.3 Nonlinear-manifold correction: the HPROM–ANN decoder

Following the general nonlinear-PMOR strategy of decomposing the ROB into a small *primary* (retained) part and a larger *secondary* (discarded) part whose amplitude is reconstructed by a trained nonlinear map – introduced for fluid-dynamics PMOR as the PROM-ANN/HPROM-ANN framework by Barnett, Farhat and Maday [17] and generalized to interpretable Gaussian-process and radial-basis-function closures by Ares De Parga et al. [18] – the total basis is partitioned as $\mathbf{V}_{\mathrm{tot}} = [\mathbf{V}\ \bar{\mathbf{V}}]$, $\mathbf{V} \in \mathbb{R}^{n_L \times n}$, $\bar{\mathbf{V}} \in \mathbb{R}^{n_L \times \bar{n}}$, $n \ll \bar{n}$, and the displacement field is approximated as

$$\boldsymbol{u}_f \approx \boldsymbol{u}_f^{\mathrm{aff}}(\boldsymbol{\varepsilon}) + \mathbf{V}\, \boldsymbol{q} + \bar{\mathbf{V}}\, \mathcal{N}(\boldsymbol{q}), \tag{4}$$

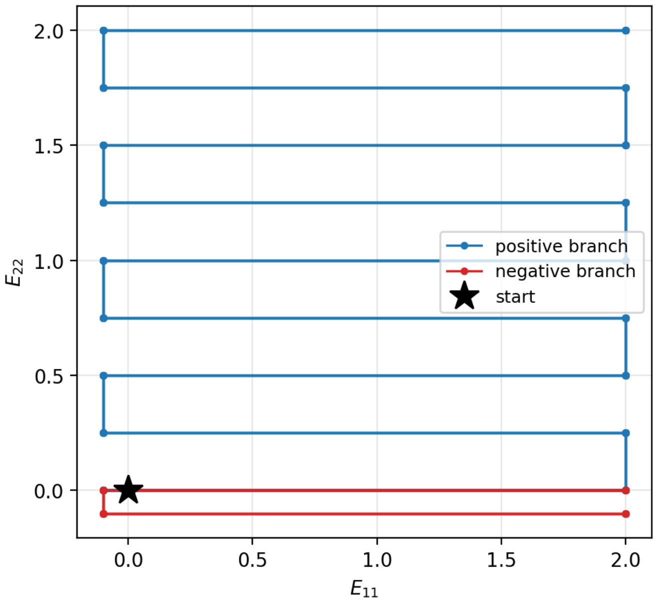

Figure 3: Stage–0/1 macro-strain training trajectories, projected onto the (E_{11}, E_{22}) plane (a third, shear component γ_{12} is also swept but omitted here for clarity). The boustrophedon (zig-zag) sweep starting from the origin covers both a positive branch ($E_{22} \in [0, 2]$) and a shorter negative branch ($E_{22} \in [-0.1, 0]$); FOM solves along these paths generate every Stage–1 training snapshot used to build the POD basis, the decoder $\mathcal{N}$, and the ECM/MAW–ECM cubature rules of this section.

with $\boldsymbol{q} \in \mathbb{R}^n$ the primary (retained) generalized coordinates and $\mathcal{N} : \mathbb{R}^n \to \mathbb{R}^{\bar{n}}$ a trained neural network that reconstructs the secondary (discarded) amplitudes. In this project, $\mathbf{V}$ and $\boldsymbol{q}$ are identified, via the same least-squares primary/secondary basis-identification procedure used by Hernández's MAW–ECM draft [19] for a different homogenization RVE, with the macro-strain $\boldsymbol{\varepsilon}$ itself – an *input-driven* manifold, valid here because the problem is static and hyperelastic (no history dependence), so the equilibrium state is uniquely parameterized by $\boldsymbol{\varepsilon}$. Consequently $\mathcal{N}$ is trained once, offline, on Stage–1 trajectories to map macro-strain directly to the reconstructed secondary coordinates, and no reduced equilibrium equation of the form (3) needs to be solved online at all if $\mathcal{N}$'s prediction is trusted directly; alternatively, (4) can still be substituted into a Galerkin-projected residual and corrected by Newton iterations, exactly as in (3). These are precisely the two HPROM–ANN operating modes evaluated in Section 15: the *iterative* mode (Newton correction to convergence) and the *direct* mode, D-HPROM–ANN (zero Newton iterations, $\mathcal{N}$'s output used as-is).

2.4 Hyperreduction: the empirical cubature method and MAW–ECM

Even after (4), evaluating the residual (1) (or the reduced residual (3)) still requires a full loop over all $n_{\text{gp}} = \sum_e n_{\text{gp}}^{(e)}$ Gauss points of the original FOM mesh, since the stress $\boldsymbol{S}_{e,g}$ must be evaluated from the true microscopic constitutive law at every one of them. Reindexing the element/Gauss-point pairs (e, g) of (1) by a single global index $g \in \{1, \dots, n_{\text{gp}}\}$ (as is standard in the ECM literature), so that $\boldsymbol{r}_g := \boldsymbol{L}_e^T \boldsymbol{B}_{e,g}^T \boldsymbol{S}_{e,g}$ and $\boldsymbol{S}_g := \boldsymbol{S}_{e,g}$ denote the residual and stress contribution of global Gauss point g, hyperreduction replaces the full Gauss-point sum by a sparse cubature rule over a small subset $\mathcal{Z} \subset \{1, \dots, n_{\text{gp}}\}$ of retained points with positive weights $\{\omega_g\}_{g \in \mathcal{Z}}$:

$$\sum_{g \in \mathcal{Z}} (\mathbf{V}^T \boldsymbol{r}_g)(\boldsymbol{u}_f; \boldsymbol{\varepsilon})\, \omega_g = \mathbf{0}, \qquad \boldsymbol{S}_{\text{macro}}(\boldsymbol{\varepsilon}) \approx \frac{1}{|\Omega_0|} \sum_{g \in \mathcal{Z}_\sigma} \boldsymbol{S}_g(\boldsymbol{u}_f; \boldsymbol{\varepsilon})\, \omega_{\sigma,g}, \tag{5}$$

one rule ($\mathcal{Z}$, weights ω_g) tailored to integrate the projected *equilibrium* residual accurately, and, in general, a second, independently constructed rule ($\mathcal{Z}_\sigma$, weights $\omega_{\sigma,g}$) tailored to the homogenized-*stress* output functional – the two point sets and weight sets are not required to coincide, and Fig. 4 shows both retained element sets for this RVE side by side.

The weights are found offline by the empirical cubature method (ECM) of Hernández, Caicedo and Ferrer [20]: a sparse, non-negative least-squares problem that selects the smallest point set capable of exactly integrating a truncated-SVD basis of the training-snapshot integrand space, in the same spirit as (and predating) the energy-conserving-sampling-and-weighting (ECSW) family used for Petrov–Galerkin hyperreduction

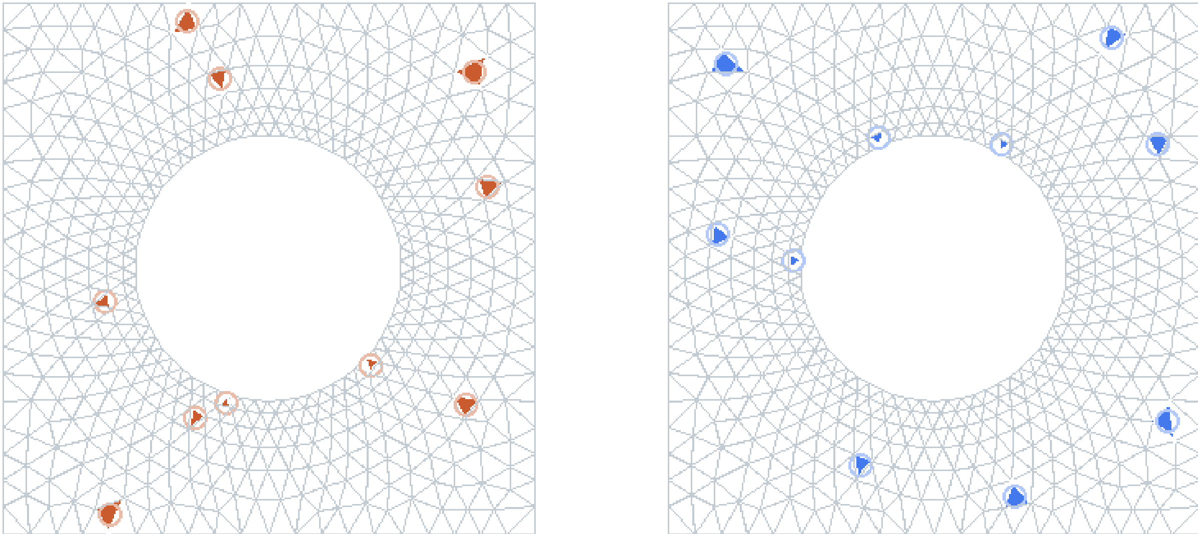

Figure 4: MAW–ECM hyperreduced integration supports for this RVE's canonical (`maw_dynamic`) configuration. Left: the $|\mathcal{Z}| = 10$ Gauss points retained for the projected equilibrium residual. Right: the $|\mathcal{Z}_\sigma| = 10$ Gauss points retained for the homogenized-stress output functional. The two supports are constructed independently and, as visible here, are not required to select the same elements.

in the fluids PMOR literature [23, 24, 25] and the empirical quadrature procedure [26, 27, 28]. A model that uses a single such *fixed* weight set for every macro-strain state is an ECM-hyperreduced HPROM; the manifold-adaptive-weight empirical cubature method (MAW–ECM) of Hernández [19], used for the weights in this paper's HPROM–ANN, instead allows ω_g (and $\omega_{\sigma,g}$) to vary smoothly with the reduced state – here, macro-strain – which lets the same support set $\mathcal{Z}$ stay accurate over a wider region of the strain path than a fixed rule could, at the cost of an offline graph-regularized pruning-and-regression stage instead of a single sparse solve. We use MAW–ECM exactly as published, with no modification, and cite it as the tool it is (see the provenance note in Section 15).

3 The admissibility checklist

This paper organizes six requirements for an admissible hyperelastic surrogate W, with $\boldsymbol{\sigma} = \partial W/\partial \boldsymbol{\varepsilon}$, into a single checklist for clarity of exposition. Each requirement is individually well established and is discussed with its own citation in the section devoted to it below; the six-item grouping itself is this paper's own organizing device, not a claim that this exact enumeration is issued by some external authority:

- **(C1) Thermodynamic consistency** (reduced Clausius–Duhem inequality): $\boldsymbol{\sigma}^T\dot{\boldsymbol{\varepsilon}} - \dot{W} \geq 0$.
- **(C2) Objectivity**: $W(\boldsymbol{\varepsilon}) = W(\boldsymbol{R}\,\boldsymbol{\varepsilon})$ for every admissible rigid rotation of the current configuration.
- **(C3) Symmetry**: $\boldsymbol{\sigma}(\boldsymbol{\varepsilon}) = \boldsymbol{\sigma}(\boldsymbol{\varepsilon})^T$.
- **(C4) Normalization**: $\boldsymbol{\sigma}(\mathbf{0}) = \mathbf{0}$ and $W(\mathbf{0}) = 0$.
- **(C5) Polyconvexity**, in the sense of Ball [1] – see Section 11.1 for its precise statement and Section 11.2 for its relation to a strictly stronger alternative condition used elsewhere in the literature.
- **(C6) Growth condition**: $W \to \infty$ as $\boldsymbol{\varepsilon} \to \infty$ (with extra terms if needed).

Sections 6–12 take each point in turn. (C1)–(C4) and (C6) are satisfied exactly, by construction, independent of training. (C5) requires the most care: Section 11 gives Ball's precise definition, the correct (weaker) necessary condition we actually falsify-test, and the explicit construction that proves our model polyconvex in Ball's sense; it also shows that a strictly stronger alternative used elsewhere in the literature is not required by polyconvexity and is, for this RVE's own ground-truth response, false. Before doing so, Section 4 places this checklist against the current physics-augmented neural network (PANN) literature: which parts are already correctly and rigorously solved elsewhere, which published claims are imprecise or documentedly incorrect, and which combination of requirements – specifically, a *general* anisotropic material with no assumed symmetry group – remains an open problem that this work targets.

4 Related work: what is proven, what is imprecise, and what remains open

This section surveys the seven works of Table 1, plus several more specific supporting references, chosen because each makes an explicit, checkable claim about one or more of (C1)–(C6) for a neural-network hyperelastic energy. Table 1 summarizes the seven-work comparison; the discussion below justifies every entry and gives the precise textual basis for each claim, since "X satisfies polyconvexity" is stated in nearly all of this literature and the differences are entirely in *which* polyconvexity-adjacent condition is actually being claimed.

Table 1: Comparison against the admissibility checklist. "Exact" means satisfied by construction for every parameter value, not approximated by training or data augmentation. "Group-free anisotropy" means the model represents a general anisotropic material without committing to a named symmetry class (isotropic, transversely isotropic, cubic, orthotropic, ...) or a learned member of one. "Growth exercised" means the model's volumetric term actually diverges/was actually tested at $J \to 0^+$, as opposed to being stated as a requirement but bypassed because the worked examples are (nearly) incompressible.

Work	Exact obj.	Exact ref. state	Ball polyconvex	Group-free anisotropy	Growth exercised
As'ad et al. [5]	yes	yes	no (E-convex)	yes (raw $\boldsymbol{E}$)	n/a
Linka & Kuhl [6]	yes	yes	yes	no (isotropic)	no (incompressible)
Tac et al. [7]	yes	partial[1]	yes	no (2 fixed fibers)	no (incompressible)
Klein et al. [4]	yes/approx.[2]	approx. only	yes	no (named group)	yes
Thakolkaran et al. [8]	yes	yes	yes	no (isotropic)	no (no barrier term)
Linden et al. [9]	yes	yes	yes	no (named group)	yes
Kalina et al. [10]	yes	yes	no	partial[3]	yes
This work	**yes**	**yes**	**yes**	**yes**	**yes**

4.1 Models that enforce a strictly stronger condition than polyconvexity

As'ad, Avery and Farhat [5] build a hyperelastic PANN for an anisotropic woven-fabric RVE homogenized with the same kind of FE^2 pipeline used in this project, and validate it inside an actual coupled fluid–structure simulation (the supersonic inflation of a parachute canopy) – the only work surveyed here that demonstrates a downstream, large-scale numerical-robustness benefit, not just held-out accuracy. Their own Section 3.3 states the reason plainly: "enforcing polyconvexity in F is difficult as E is not in general convex in F. On the other hand, consider the convexity of W in E, which is equivalent to... $\mathbb{C} = \partial \boldsymbol{S}/\partial \boldsymbol{E} \succeq 0$." This is, verbatim, the SPD-tangent condition discussed in Section 11.2: a global convex ICNN is applied directly to the raw Green–Lagrange strain, which by elementary calculus forces $\partial \boldsymbol{S}/\partial \boldsymbol{E} \succeq 0$ *everywhere*, not merely at the reference state. The authors are explicit that this is a deliberate substitute for polyconvexity, not an oversight, and they give a real justification for it: their own Eq. (13) decomposes the F-Hessian of W into a material-stiffness term (made non-negative by convexity in E) and a geometric-stiffness term built from the ambient stress $\boldsymbol{S}$ itself, which can be negative in compression – so ellipticity in F (and hence buckling-type non-uniqueness) can still be lost even though $\partial \boldsymbol{S}/\partial \boldsymbol{E} \succeq 0$ holds globally. This is a legitimate and carefully reasoned engineering trade-off, and their finite-element examples (Section 4.4) show it buys real numerical robustness: Newton's method diverges for a non-convex ANN under load, and converges for their convex-in-$\boldsymbol{E}$ one.

It is nonetheless exactly the condition Section 11.2 argues against for a material like the one in this project. Unlike convexity in F itself – which As'ad et al.'s own Section 3.3 correctly notes is incompatible with objectivity, citing Coleman and Noll's classical counterexample – convexity in $\boldsymbol{E}$ raises no such contradiction, since $\boldsymbol{E}$ (equivalently C) is already an objective strain measure; Gao, Neff, Roventa and Thiel [11] build a valid existence theory on exactly this (stronger-than-polyconvex) hypothesis, so it is a coherent, usable condition in general, not an impossible one. The reason to avoid it for this RVE is not a general impossibility but a specific, measured fact: the FOM tangent audit gives a negative eigenvalue, $-0.0648\,\mathrm{GPa}$, at a Stage–10 state, so $\partial \boldsymbol{S}/\partial \boldsymbol{E} \succeq 0$ is simply false for this material's true response, and a convex-in-$\boldsymbol{E}$ ICNN applied to this RVE would be structurally unable to reproduce that state. Two further, more mechanical differences follow directly from this choice. First, As'ad et al.'s stress-free-reference correction is a fully general linear term $\boldsymbol{h} \cdot \boldsymbol{E}$ with $\boldsymbol{h} = -\partial\theta/\partial \boldsymbol{E}(\boldsymbol{0})$ (their Eqs. 15–16): adding *any* linear function of $\boldsymbol{E}$ preserves convexity in $\boldsymbol{E}$, so

[1] Exact for their CANN/NODE cores; an unaddressed gap for their ICNN core, whose biases are unconstrained.

[2] Exact for their invariant model W^I; only approximated by data augmentation for their deformation-gradient model W^F.

[3] Anisotropy *parameters* (e.g. fiber directions) within a pre-chosen symmetry class are learned; the class itself is still user-specified.

no special structure among the three strain components is required. A linear-in-$\boldsymbol{E}$ correction is a *quadratic*, generally non-convex-in-F form once expressed in F, so this shortcut is not available to a polyconvex-in-F construction; this is precisely why Proposition 2 needs the balanced-direction identity (21) instead of a free 3-parameter correction. Second, As'ad et al.'s ICNN needs only the classical Amos et al. convexity-only construction (unconstrained input-to-hidden weights), because it is applied directly to the raw strain; our construction needs the strictly stronger *monotone* ICNN (non-negative weights at every layer, Section 11.4) because the ICNN here is composed with an intermediate layer of engineered convex features of F, and lifting convexity through that composition requires the outer function to be coordinate-wise non-decreasing. Finally, their default loss normalizes every stress component by its own per-component standard deviation (their Eq. 2) – i.e. always at the extreme $0.00 = 1$ of (25) – which Table 5 shows costs a factor of roughly two in von Mises accuracy for an RVE whose components differ in magnitude by two orders of magnitude; this trade-off is not discussed in [5], most likely because their fabric material's stress components do not differ as severely.

The same global-SPD-tangent condition, by a different mechanism, underlies Xu, Huang and Darve [12]: rather than an ICNN, their SPD-NN predicts the Cholesky factor of the tangent stiffness directly, so that the stiffness matrix is symmetric positive definite by construction and stress is integrated from it incrementally. This is applied to anisotropic fiber-reinforced plates and to path-dependent (elasto-plastic) response, which the ICNN constructions surveyed here do not attempt, at the cost of the same strictly-stronger-than-polyconvex commitment as Section 4.1, for the same reason: a positive-definite tangent stiffness *is* $\boldsymbol{D} \succeq 0$, restated in rate form.

4.2 Models that enforce Ball's polyconvexity, but only for a named symmetry group

Klein et al. [4] (already used throughout Section 11 as the primary reference for the correct definition) give two ICNN constructions, both restricted to a material symmetry group $\mathcal{G} \subset SO(3)$ specified a priori (cubic, transversely isotropic) via a structural tensor. Their invariant-based model W^I gets objectivity, material symmetry, and polyconvexity exactly, but not the reference state: "the model (16) is not stress-free by construction . . . this may thus lead to a loss of ellipticity." Their own explanation for why is worth quoting in full, because it is the state of the art this project improves on: "In Fernández et al. (2021b), a projection approach is proposed which fulfills the stress-free reference configuration in an exact way. While this approach preserves polyconvexity when formulated in terms of F instead of C, it is not compatible with the methods of incorporating objectivity and material symmetry which are used in this work, and therefore cannot be applied." Their deformation-gradient model W^F makes the opposite trade: exact polyconvexity and material symmetry, but objectivity is only *approximated* by data augmentation over 1024 random rotations (their Eq. 22), which they candidly report requires 64 observers before the shear response becomes acceptably tight (their Table 1, Fig. 4).

Linden et al. [9] close this specific gap for isotropic and transversely isotropic materials: they construct an additive stress- and energy-normalization term, restricted to J and the symmetry-respecting anisotropic invariants (never the raw off-diagonal components of C), and show this achieves exact objectivity, exact reference state, exact polyconvexity, and exact material symmetry *simultaneously* – solving the open problem Klein et al. identified. Critically, however, they do this only within a symmetry class fixed by a structural tensor, and they are explicit that this does not extend to a general material: "there are symmetry groups for which no complete set can be found that does not violate polyconvexity . . . it might be possible that there does not exist a set of invariants to describe the anisotropy of materials with arbitrary microstructures, e.g., 3D-printed composites." This is, in the words of the researchers who solved the harder half of this problem, an open question for exactly the case this paper addresses: an RVE with no known or assumed symmetry class. The mechanism used here answers it for that case: the balanced, non-D_4-closed direction systems of Table 4 are not tied to any structural tensor or symmetry group, and Proposition 1 shows they still produce the isotropic reference-derivative structure that an exact, polyconvexity-preserving correction requires.

A very recent preprint by an overlapping set of authors, Klein et al. [35], extends Linden et al.'s exact-triple result (exact objectivity, exact reference state, exact polyconvexity, all simultaneous) from isotropic/transversely-isotropic materials to a substantially wider set of named finite symmetry groups (cubic, tetragonal, and others), using a triclinic (fully general) integrity basis of C, $\mathrm{cof}(C)$, and J that is subsequently restricted to a chosen group $\mathcal{G}$ by an explicit group-symmetrization operator. This is real, recent progress on the specific gap this paper also targets, and we have checked it directly rather than taking the title at face value: their own construction still requires $\mathcal{G}$ to be selected a priori for every worked example, and the paper's own abstract, conclusion, and Section 4.3.2 scope the contribution to "finite symmetry groups" throughout, with a single undeveloped aside noting the construction "could also be employed without incorporating material symmetry at all" – a possibility they neither formalize nor verify against any of (C1)–(C3). The gap

Linden et al. identified for a material with no known or assumed symmetry class therefore remains open even after this update, and the mechanism of this paper (the balanced, non-D_4-closed direction systems above) is, to our knowledge as of this writing, still the only construction that addresses it.

4.3 A model that widens the anisotropy class by giving up polyconvexity

Kalina et al. [10] take the opposite trade from Section 4.2: they still require a user-specified symmetry class (one of 14 crystallographic groups), but the structure tensor's own parameters (e.g. fiber directions) within that class are learned jointly with the network from data, which is a substantially more flexible anisotropy treatment than a fixed structural tensor. In exchange, they explicitly decline polyconvexity: "polyconvexity can be too restrictive, especially for multiscale modeling," relying instead on physics-augmented correction terms in the loss rather than an architecturally convex core. Together with Section 4.2, this is the clearest evidence in the current literature for a real trade-off frontier: the more general the anisotropy representation gets, the harder (and, so far, unachieved outside this work) it has been to keep an exact, architectural polyconvexity certificate.

4.4 Isotropic and fiber-restricted polyconvex CANNs, and the growth condition in practice

Linka & Kuhl [6], Tac et al. [7], and Thakolkaran et al. [8] all correctly use the additive, block-decoupled sufficient condition for polyconvexity (convex, non-decreasing scalar functions of I_1, of a polyconvex modification of I_2, and of J, summed), and Tac et al. are unusually careful to state that this is strictly stronger than what is needed and to distinguish it explicitly from rank-one convexity: "[r]ank-one convexity [is] weaker than polyconvexity and not sufficient for the existence of global minimizers." None of the three, however, is anisotropic beyond at most two fixed fiber directions (transverse isotropy), and the growth condition (C6) is uniformly *stated but not exercised*: Linka & Kuhl and Tac et al. both work with perfectly or nearly incompressible materials, so $J = 1$ is enforced by a Lagrange-multiplier pressure and the stated $J \to 0^+$ divergence is never actually active in a trained model. Thakolkaran et al.'s ICKAN paper is, on the polyconvexity definition itself, the most careful of the three – it states Ball's condition precisely and does not conflate it with anything stronger – but it uses $K_3 = (J-1)^2$ as its sole volumetric term, which is finite (equal to 1) at $J = 0$: there is no divergent barrier at all, and (C6) is not satisfied by their model as published. Our own volumetric term is exercised, not merely stated: the audit in Table 3–4 and Figure 5 samples $W_{\rm pc}$ down to $J = 10^{-6}$ and up to $J = 100$ for the trained model, for a compressible, general anisotropic material, which is a more demanding setting than any of the three incompressible or bounded-volumetric examples above.

Two further isotropic-only works round out this picture, at opposite ends of the convexity-vs-flexibility trade-off already seen in Section 4.3. Tac, Sahli-Costabal and Buganza Tepole [13] originate the neural-ODE route to the same additive, monotone-in-the-invariants sufficient condition used in [6, 7], demonstrated on anisotropic (two-fixed-fiber) skin data but, like the others in this subsection, without an exercised $J \to 0^+$ barrier. Abdolazizi, Aydin, Cyron and Linka [14], by contrast, apply a plain, unconstrained Kolmogorov–Arnold network to the same isotropic invariants and explicitly decline any convexity requirement at all: objectivity, material symmetry, and the reference state are enforced by construction, but polyconvexity, rank-one convexity, and growth are not discussed as design goals anywhere in the paper. Their examples (rubber, brain tissue, silicone) show good fits with no reported instability, which is consistent with, not evidence against, the point made throughout this section: losing the polyconvexity certificate does not necessarily show up as a visible failure on in-distribution calibration data, which is precisely why it needs to be guaranteed by construction rather than checked post hoc.

4.5 Where published claims have been documented to be wrong

Beyond the SPD/polyconvexity distinction drawn in Section 11.2, the literature itself documents concrete, checkable errors of exactly this kind, which is useful context for how easy this is to get wrong even with a stated intent to be rigorous. Klein et al. [4] report that Ghaderi et al. [15] claim a polyconvex ANN potential but use hyperbolic-tangent activations, which are not convex, so "the potential proposed in Ghaderi et al. (2020) violates the polyconvexity condition." We checked this directly against the original rather than taking it on trust: Ghaderi et al. do state, in their own Appendix B.1, that "[i]f weights, which connect the input of λ_j to other neurons, are positive, the proposed model holds the condition of polyconvexity," and their Section 4 specifies the network core uses "soft plus . . . , sinusoid . . . and hyperbolic tangent" activations together. Weight non-negativity alone does not make a composition convex if the activation itself is not convex and non-decreasing (Section 11.4); sine is neither, and tanh is concave, not convex, on the positive axis. Klein et al.'s report is accurate, and the error is exactly the kind Section 11.4 is built to avoid by construction: stating a sufficient condition on the weights while leaving out that it also requires convex, non-decreasing activations. They report the same failure mode for Tac et al. (2021), whose model uses the invariant $\bar{I}_2 = \frac{1}{2}(\operatorname{tr}\bar{C})^2 - \frac{1}{2}\operatorname{tr}\bar{C}^2$ as a direct network input: this particular invariant is *not* polyconvex (Hartmann and Neff, cited by Klein et al. as their Lemma 2.4), so despite the stated intent, "the polyconvexity condition is violated by the choice of input arguments." Both are examples of the same lesson emphasized in Section 11.4: polyconvexity is a property of the specific functional form all the way down to the choice of activation function and input invariant, not a label that can be attached to an architecture in general, and a single non-convex activation or a single non-polyconvex invariant among otherwise-correct ones is enough to void the entire certificate.

4.6 What remains open, and what this work contributes

Reading Sections 4.1–4.5 together, the state of the art as of this writing has, to our knowledge, not produced a model that is simultaneously (i) exactly objective, (ii) exactly stress-free and energy-free at the reference state, (iii) exactly polyconvex in Ball's sense, and (iv) anisotropic without committing to a named symmetry group – the combination Linden et al. [9] explicitly flag as possibly requiring an invariant set that does not exist for a general microstructure. This paper's contribution is a specific, checkable answer for the case of a 2D RVE: replace the symmetry-group structural tensor with several *generic*, non-D_4- closed direction systems, each individually balanced (Table 4), so that Proposition 1's isotropic-reference-derivative argument goes through without ever assuming or checking which symmetry class, if any, the material belongs to. A secondary, smaller contribution is the loss-design result of Section 13: none of the seven works surveyed above discusses the possibility that per-component loss normalization (used, in its most extreme form, by [5], and in a milder blended form by 0.65 of the previously certified version of this model) can measurably *hurt* accuracy on the physically relevant equivalent stress when the material's stress components differ by orders of magnitude, purely because it forces capacity to be spent chasing the smallest, noisiest component's relative error. Finally, a limitation this work shares with every polyconvex-PANN paper surveyed except As'ad et al.: none of Sections 6–15 demonstrate that the polyconvexity certificate translates into the kind of downstream finite-element robustness (guaranteed Newton convergence, bounded transient kinetic energy) that [5] demonstrates for their strictly stronger, E-convex construction. Embedding this certified polyconvex PANN in an actual structural or multiscale solve, and checking whether the same robustness benefits appear without requiring the stronger and, for this material, false SPD-tangent assumption, is the natural next step to make this defensible end-to-end rather than only at the level of the standalone constitutive surrogate.

5 Kinematics and stress convention

Let F be the deformation gradient, and

$$C = F^T F, \qquad E = \tfrac{1}{2}(C - I), \qquad J = \det F = \sqrt{\det C} > 0. \tag{6}$$

The database and the model use the engineering-shear coordinate

$$\boldsymbol{\varepsilon} = \begin{bmatrix} E_{11} \\ E_{22} \\ \gamma_{12} \end{bmatrix}, \qquad \gamma_{12} = 2E_{12}, \tag{7}$$

so that the work identity

$$dW = S_{11}\, dE_{11} + S_{22}\, dE_{22} + S_{12}\, d\gamma_{12}, \qquad \boldsymbol{\sigma} := \frac{\partial W}{\partial \boldsymbol{\varepsilon}} = \begin{bmatrix} S_{11} \\ S_{22} \\ S_{12} \end{bmatrix} \tag{8}$$

returns exactly the second Piola–Kirchhoff components stored in the direct FOM data, with no factor-two ambiguity in the shear term and no independently trained stress network: every stress reported in this document is $\partial W/\partial \boldsymbol{\varepsilon}$ of the same scalar W.

6 (C1) Thermodynamic consistency

For a purely elastic RVE with no internal (dissipative) state variables, the reduced isothermal Clausius–Duhem inequality is $\mathcal{D} := \boldsymbol{\sigma}^T \dot{\boldsymbol{\varepsilon}} - \dot{W} \geq 0$. Since $W = W(\boldsymbol{\varepsilon}(t))$ and W is differentiable,

$$\dot{W} = \frac{\partial W}{\partial \boldsymbol{\varepsilon}} \cdot \dot{\boldsymbol{\varepsilon}} = \boldsymbol{\sigma}^T \dot{\boldsymbol{\varepsilon}} \tag{9}$$

identically, for *any* smooth scalar potential W and *any* strain history $\boldsymbol{\varepsilon}(t)$, by the chain rule alone. Hence $\mathcal{D} \equiv 0 \geq 0$: the inequality holds with equality, and the model is thermodynamically consistent (fully reversible, zero dissipation) for the same reason every hyperelastic (Green-elastic) model is – because the stress is defined as $\partial W/\partial \boldsymbol{\varepsilon}$ in (8) and not as an independently parameterized function. This is not a property that training can violate or that needs to be checked numerically; it is a property of the functional form (8) itself.

7 A pure-regression baseline: violating (C1) directly

Section 6 shows (C1) holds for *any* model of the form $\boldsymbol{S} = \partial W/\partial \boldsymbol{\varepsilon}$, regardless of which W is learned. It says nothing about a model that is not of this form. As'ad et al. [5] motivate their own hyperelastic-ANN construction by first exhibiting exactly such a model – a plain regression network that maps strain directly to stress, with no potential anywhere – as their baseline "regression ANN", the bottom tier of a three-tier ablation (regression ANN, hyperelastic ANN, convex hyperelastic ANN) that motivates each of their architectural commitments in turn. This section repeats that ablation for this project's own RVE and its own admissibility checklist, adding a fourth tier below the two certified polyconvex cores and the uncertified free baseline of Section 17:

- **Tier 1 (this section)**: a pure-regression MLP, widths $(128, 128, 64)$, mapping normalized $[E_{11}, E_{22}, \gamma_{12}]$ straight to normalized $[S_{11}, S_{22}, S_{12}]$ – `stress()` is a forward pass, never a gradient. No energy functional exists in this model at all.
- **Tier 2**: the free hyperelastic PANN of Section 17 – an unconstrained energy W_θ, so (C1) holds, but nothing else in (C4)–(C6) is guaranteed.
- **Tiers 3a/3b**: the certified polyconvex ICNN and ICKAN of Sections 11.4 and 16, satisfying all six checklist items exactly.

A direct, quantitative test of (C1). For a genuine hyperelastic model, integrating $\dot{W} = \boldsymbol{S} \cdot \dot{\boldsymbol{\varepsilon}}$ (Section 6) along any closed strain loop $\mathcal{L}$ – a path in strain space with the same start and end point – gives

$$\oint_{\mathcal{L}} \boldsymbol{S} \cdot d\boldsymbol{\varepsilon} = \oint_{\mathcal{L}} dW = W(\boldsymbol{\varepsilon}_{\text{end}}) - W(\boldsymbol{\varepsilon}_{\text{start}}) = 0, \tag{10}$$

exactly, for *every* closed loop, independently of which admissible W produced $\boldsymbol{S}$: this is a direct corollary of (C1), not a new physical assumption. For the tier-1 pure-regression model, $\boldsymbol{S}$ is not the gradient of anything, so nothing in its construction forces (10), and there is no reason to expect it numerically. We evaluate (10) by trapezoidal quadrature (200 points per edge) on four closed polygonal loops entirely inside the Stage–1 training range (two axis-aligned normal-strain squares of different size, a shear loop, and an irregular triangle mixing normal and shear strain), for all four tiers, using each model's own stress output (a direct forward pass for tier 1, $\partial W_\theta/\partial \boldsymbol{\varepsilon}$ by automatic differentiation for tiers 2–3b).

Table 2: Cyclic work $\oint_{\mathcal{L}} \boldsymbol{S} \cdot d\boldsymbol{\varepsilon}$ (Pa) around four closed strain loops, trapezoidal quadrature, 200 points per edge. Exactly zero is expected, by (10), for tiers 2–3b; nothing constrains tier 1.

Loop	Pure regression (tier 1)	Free (tier 2)	ICNN (tier 3a)	ICKAN (tier 3b)
Small normal square	$-3.042e+04$ Pa	$3.024e+00$ Pa	$-1.490e-08$ Pa	$0.000e+00$ Pa
Large normal square	$-6.687e+04$ Pa	$4.128e+00$ Pa	$0.000e+00$ Pa	$1.192e-07$ Pa
Shear loop	$-3.583e+04$ Pa	$8.769e+00$ Pa	$-2.757e-07$ Pa	$7.451e-09$ Pa
Mixed triangle	$9.850e+03$ Pa	$-4.771e+01$ Pa	$-4.968e+01$ Pa	$-4.922e+01$ Pa

For context, a typical energy density for this RVE (the training-set energy scale used to normalize the loss) is $2.136e+09$ Pa: the tier-1 residual on the two normal-strain loops and the shear loop is five to six orders of magnitude smaller than this scale (0.003% of it at worst) – small in absolute terms, but, as the next paragraph shows, not a quadrature artifact, unlike the superficially similar-looking residuals of tiers 2–3b on the mixed triangle loop.

Distinguishing a real violation from quadrature error. The mixed-triangle residuals of tiers 2–3b in Table 2 are not exactly zero at 200 points per edge, because the trapezoidal rule only approximates a line integral of a nonlinear gradient field; but (10) holds exactly in the continuum limit, so this residual must vanish as the quadrature is refined, regardless of which admissible W produced $\boldsymbol{S}$. The tier-1 residual has no such reason to vanish, because it is not quadrature error in the first place – it is the trapezoidal estimate of a line integral of a non-conservative vector field, which is generally path-dependent and nonzero even in the continuum limit. Refining the same *mixedtriangle* loop by $4\times$ and $16\times$ ($200, 800, 3200$ points per edge) confirms exactly this:

- Pure regression (tier 1): $9.850e+03\,\mathrm{Pa}, 9.900e+03\,\mathrm{Pa}, 9.902e+03\,\mathrm{Pa}$ – stable, non-vanishing.
- Free (tier 2): $-4.771e+01\,\mathrm{Pa}, -3.160e+00\,\mathrm{Pa}, 9.079e-02\,\mathrm{Pa}$ – shrinking toward zero.
- ICNN (tier 3a): $-4.968e+01\,\mathrm{Pa}, -3.105e+00\,\mathrm{Pa}, -1.940e-01\,\mathrm{Pa}$ – shrinking toward zero.
- ICKAN (tier 3b): $-4.922e+01\,\mathrm{Pa}, -3.075e+00\,\mathrm{Pa}, -1.921e-01\,\mathrm{Pa}$ – shrinking toward zero.

The three energy-based tiers converge to zero at the rate expected of trapezoidal quadrature error for a smooth integrand ($\mathcal{O}(h^2)$, i.e. roughly a $16\times$ reduction per $4\times$ refinement); the tier-1 residual instead converges to a fixed nonzero constant. This is not a difference of degree: it is the signature of a genuine curl in the tier-1 stress field, i.e. a direct, measured violation of (C1), as distinct from finite-sample noise.

Accuracy does not reveal this violation – if anything, it hides it. Table 6 reports the tier-1 model's own held-out Stage–10 accuracy: 0.054% overall stress error and 0.033% von Mises error, *better than every other tier in this study*, including both certified polyconvex cores. This is the sharpest version yet of the point made in Section 17 with the free baseline: an aggregate accuracy metric on a held-out trajectory not only fails to detect the absence of a physical constraint, it can reward the model that discards the most physics, because a direct regression has no architectural constraint competing with the stress fit at all. The cyclic-work test of Table 2 is not an alternative way of measuring the same thing accuracy already measures; it is the only test in this paper that checks (C1) itself; direct observation, not a metric to be simultaneously optimized.

8 (C2) Objectivity

Objectivity concerns a superposed rigid rotation $Q \in \mathrm{SO}(2)$ of the *current* configuration, $F \mapsto QF$. Since

$$(QF)^T(QF) = F^TQ^TQF = F^TF = C, \tag{11}$$

any scalar built only from C (equivalently, from F^TF, $\mathrm{cof}(F)^T\,\mathrm{cof}(F) = \mathrm{cof}(C)$, and $\det F$, as in Section 11) automatically satisfies $W(QF) = W(F)$ for every $Q \in \mathrm{SO}(2)$. Every feature used by the model in Section 11 is built exclusively from C and $\mathrm{cof}(C)$, so objectivity holds identically, not approximately, and not because it was included as a training penalty. This is different from *material* symmetry, which concerns rotating the RVE relative to its own fixed material axes and is discussed next.

9 (C3) Symmetry: two different meanings

The word "symmetry" is used for two unrelated properties in continuum mechanics, and the checklist item (C3) refers to the first, not the second:

- **Tensor symmetry**, $\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$: a basic consequence of balance of angular momentum in the absence of body couples. It must hold for *any* admissible constitutive model, including anisotropic ones.
- **Material symmetry**: invariance of W under a *material* symmetry group (e.g. the D_4 group of a square lattice), $W(C_{11}, C_{22}, C_{12}) = W(C_{22}, C_{11}, -C_{12})$. This is a *modelling choice*, not a conservation law, and this project deliberately does *not* impose it: the RVE is treated as a general anisotropic material, because a square-shaped numerical cell does not by itself make the effective homogenized material isotropic or D_4 symmetric.

Tensor symmetry (the actual content of C3) holds identically in this model: the second Piola stress is parameterized by exactly the three independent numbers (S_{11}, S_{22}, S_{12}) of (8), which are then placed into the symmetric matrix $\left[\begin{smallmatrix} S_{11} & S_{12} \\ S_{12} & S_{22} \end{smallmatrix}\right]$. There is no separate, independently learned S_{21} that could disagree with S_{12}: $\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$ by the structure of the parameterization, not as an approximate or penalized identity. Material symmetry (D_4) is a different, optional property that this model does not claim.

10 (C4) Normalization: the stress-free reference state

10.1 The energy value: trivial by construction

Let Φ_θ be a scalar network of a feature vector $\boldsymbol{z}_{\mathrm{pc}}(F)$ defined in Section 11, and let $\boldsymbol{z}_{\mathrm{pc}}(I)$ denote the features evaluated at the reference state $F = I$. The energy is defined as

$$W_{\mathrm{pc}}(F) = \Phi_\theta\big(\boldsymbol{z}_{\mathrm{pc}}(F)\big) - \Phi_\theta\big(\boldsymbol{z}_{\mathrm{pc}}(I)\big) - r\log J + \frac{\beta}{2}(J-1)^2. \tag{12}$$

At $F = I$: $\boldsymbol{z}_{\mathrm{pc}}(F) = \boldsymbol{z}_{\mathrm{pc}}(I)$ so the first two terms cancel exactly; $J = 1$ so $\log J = 0$ and $(J-1)^2 = 0$. Hence $W_{\mathrm{pc}}(I) = 0$ *exactly*, for *any* network Φ_θ and *any* parameters θ, before any training happens. This part of (C4) requires no special argument about the structural directions.

10.2 The stress value: not trivial, and this is where the balanced directions matter

Differentiating (12) and evaluating at $\boldsymbol{\varepsilon} = 0$ is not automatically zero: it requires the gradient of the Φ_θ term to be exactly cancelled by the $-r\log J$ term, which only has an isotropic (equal-normal, zero-shear) shape. We now show this cancellation in full.

Lemma 1 (Isotropic reference derivative)**.** *Let d be a unit vector in $\mathbb{R}^2$. Then* $\operatorname{cof}(d\otimes d) = I - d\otimes d$.

Proof. For $d = (\cos\theta, \sin\theta)$, $d\otimes d = \begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}$, and by the 2×2 cofactor formula $\operatorname{cof}(d\otimes d) = \begin{bmatrix} \sin^2\theta & -\cos\theta\sin\theta \\ -\cos\theta\sin\theta & \cos^2\theta \end{bmatrix} = I - d\otimes d$, using $\cos^2\theta + \sin^2\theta = 1$. □

Proposition 1 (Every feature has an isotropic C-gradient at $C = I$)**.** *Let d_{k1}, d_{k2}, d_{k3} be unit vectors and $w_{k1}, w_{k2}, w_{k3} > 0$ satisfy the balance condition $\sum_i w_{ki}\, d_{ki}\otimes d_{ki} = I$ for each system $k = 1,2,3$, and set (Section 11)*

$$I_1 = \operatorname{tr} C, \quad Q^F_{k,p} = \sum_i w_{ki}(d_{ki}\cdot C d_{ki})^p, \quad Q^H_{k,p} = \sum_i w_{ki}(d_{ki}\cdot \operatorname{cof}(C) d_{ki})^p, \quad p = 2,3.$$

Then at $C = I$, every one of these $1 + 12 = 13$ scalar features, together with J and J^2, has a C-gradient proportional to the identity matrix, with the explicit coefficients in Table 3.

Proof. Taking the trace of the balance identity, $\sum_i w_{ki}|d_{ki}|^2 = \operatorname{tr}(I) = 2$ and $|d_{ki}| = 1$ gives $\sum_i w_{ki} = 2$ for every system k (matching the numerical weights recorded in Table 4).

I_1: $dI_1/dC = I$ exactly, for every C.

$Q^F_{k,p}$: $dQ^F_{k,p}/dC = \sum_i p\, w_{ki}(d_{ki}\cdot C d_{ki})^{p-1}\, d_{ki}\otimes d_{ki}$. At $C = I$, $d_{ki}\cdot I d_{ki} = |d_{ki}|^2 = 1$, so $dQ^F_{k,p}/dC|_{C=I} = p\sum_i w_{ki}\, d_{ki}\otimes d_{ki} = p\, I$.

$Q^H_{k,p}$: since $d\cdot\operatorname{cof}(C)d = \operatorname{cof}(C) : (d\otimes d)$ and the cofactor operation is self-adjoint under the Frobenius product for 2×2 matrices ($A : \operatorname{cof}(B) = B : \operatorname{cof}(A)$, verified by direct expansion), $d\cdot\operatorname{cof}(C)d = C : \operatorname{cof}(d\otimes d)$, which is *linear* in C with constant gradient $d(d\cdot\operatorname{cof}(C)d)/dC = \operatorname{cof}(d\otimes d) = I - d\otimes d$ by Lemma 1. Hence $dQ^H_{k,p}/dC = \sum_i p\, w_{ki}(d_{ki}\cdot\operatorname{cof}(C)d_{ki})^{p-1}(I - d_{ki}\otimes d_{ki})$. At $C = I$, $\operatorname{cof}(I) = I$ so $d_{ki}\cdot\operatorname{cof}(I)d_{ki} = 1$, giving $dQ^H_{k,p}/dC|_{C=I} = p\big[(\sum_i w_{ki})I - \sum_i w_{ki} d_{ki}\otimes d_{ki}\big] = p(2I - I) = p\, I$, using $\sum_i w_{ki} = 2$ and the balance identity again.

J, J^2: $dJ/dC = \frac{1}{2J}\operatorname{cof}(C)$ (standard determinant derivative), so $dJ/dC|_{C=I} = \frac{1}{2}I$; then $dJ^2/dC = 2J\, dJ/dC = \operatorname{cof}(C)$, so $dJ^2/dC|_{C=I} = I$. □

Table 3: Isotropic-derivative coefficient c_m (i.e. $dz_m/dC|_{C=I} = c_m\, I$) for each of the 15 features, from Proposition 1.

Feature	I_1	$Q^F_{k,2}$, $Q^H_{k,2}$ (each of 3 systems)	$Q^F_{k,3}$, $Q^H_{k,3}$ (each of 3 systems)	J	J^2
c_m	1	2	3	1/2	1

Proposition 2 (Exact cancellation of the reference stress)**.** *Define*

$$r := \frac{1}{2}\left(\frac{\partial\Phi_\theta}{\partial E_{11}} + \frac{\partial\Phi_\theta}{\partial E_{22}}\right)\bigg|_{\boldsymbol{\varepsilon}=\mathbf{0}}. \tag{13}$$

Then $\boldsymbol{\sigma}(\mathbf{0}) = \mathbf{0}$ exactly.

Proof. By the chain rule and Proposition 1,

$$\frac{d\Phi_\theta}{dC}\bigg|_{C=I} = \sum_m \frac{\partial\Phi_\theta}{\partial z_m}\bigg|_{\boldsymbol{z}(I)} \frac{dz_m}{dC}\bigg|_{C=I} = \Big(\sum_m c_m \frac{\partial\Phi_\theta}{\partial z_m}\bigg|_{\boldsymbol{z}(I)}\Big) I =: \lambda I. \tag{14}$$

Because Φ_θ is an ICNN with non-negative weights at every layer (Section 11), $\partial\Phi_\theta/\partial z_m \geq 0$ for every feature m; combined with $c_m > 0$ from Table 3, $\lambda \geq 0$ automatically (a non-negative combination of non-negative terms), independent of training. Using $C = I + 2E$, the chain rule gives $d\Phi_\theta/dE|_{E=0} = 2\, d\Phi_\theta/dC|_{C=I} = 2\lambda I$, i.e. in the coordinate (7),

$$\frac{\partial\Phi_\theta}{\partial\boldsymbol{\varepsilon}}\bigg|_{\mathbf{0}} = (2\lambda,\ 2\lambda,\ 0)^T, \tag{15}$$

because the off-diagonal (γ_{12}) entry of $2\lambda I$ is zero. Meanwhile $\log J = \frac{1}{2}\log\det C$ gives $d(\log J)/dC = \frac{1}{2\det C}\operatorname{cof}(C)$, so $d(\log J)/dC|_{C=I} = \frac{1}{2}I$ and, again using $C = I + 2E$, $\partial(\log J)/\partial\boldsymbol{\varepsilon}|_{\mathbf{0}} = (1, 1, 0)^T$. By definition (13), $r = \frac{1}{2}(2\lambda + 2\lambda) = 2\lambda$, so that

$$\boldsymbol{\sigma}(\mathbf{0}) = \underbrace{(2\lambda, 2\lambda, 0)^T}_{\text{from } \Phi_\theta} - r\ \underbrace{(1, 1, 0)^T}_{\text{from } -r\log J} + \underbrace{\beta(J-1)\nabla J|_{J=1}}_{=0,\text{ since } J=1} = \mathbf{0}. \tag{16}$$

□

Remark 1. *The balance condition* $\sum_i w_{ki} d_{ki} \otimes d_{ki} = I$ *is what makes this possible with a* single *scalar r: without it, the reference gradient of* Φ_θ *would generally be an anisotropic* 3*-vector (unequal normal components, nonzero shear), and no scalar multiple of the purely isotropic direction* $(1, 1, 0)^T$ *could cancel it exactly. The construction would then leave a spurious residual stress at the undeformed configuration – a real defect, unlike the SPD-tangent distinction drawn in Section 11, which does not indicate any defect of the FOM at all.*

11 (C5) Polyconvexity

11.1 Definition

Polyconvexity (Ball [1]) is a condition on the *functional form* of W as a function of F, namely that there exists a jointly convex function G of three *independent* block arguments such that

$$W(F) = G(F, \operatorname{cof} F, \det F), \qquad G \text{ convex on } \mathbb{R}^{2\times 2} \times \mathbb{R}^{2\times 2} \times \mathbb{R}. \tag{17}$$

This is an existence-theory condition: Ball used it (together with suitable growth, Section 12) to prove, via the direct method of the calculus of variations, that minimizers of the total elastic energy exist. It says nothing by itself about the sign of $\partial\boldsymbol{\sigma}/\partial\boldsymbol{\varepsilon}$ at a given deformation state. The correct hierarchy (Ball [1]; see also Dacorogna's standard reference on the calculus of variations) is

$$\text{convex} \implies \text{polyconvex} \implies \text{quasiconvex} \implies \text{rank-one convex}, \tag{18}$$

with each arrow strict in general; Section 11.4 proves our construction satisfies (17) directly, and Section 11.3 gives the weakest, pointwise-checkable necessary consequence of it that we verify numerically.

11.2 A strictly stronger alternative used elsewhere, and why it is not adopted here

Polyconvexity (17) is not the only architectural route to a well-posed hyperelastic model, and it is worth being precise about how it relates to a different, strictly stronger condition used elsewhere in the literature: a globally symmetric positive definite (SPD) tangent, $\boldsymbol{D} = \partial\boldsymbol{\sigma}/\partial\boldsymbol{\varepsilon} \succ 0$ everywhere. As discussed in Section 4.1, As'ad, Avery and Farhat [5] adopt exactly this stronger condition for their own anisotropic fabric RVE, for a specific, well-documented reason (Newton-solver robustness under load, verified in their own coupled fluid–structure simulations); Gao, Neff, Roventa and Thiel [11] likewise build a valid global-minimizer existence theorem directly on convexity of W in C, which the same reparametrization $C = I + 2E$ shows is equivalent to this SPD-tangent condition. Both are legitimate, coherent design choices, not errors – but neither is polyconvexity: Ball's existence theorem needs only (17) together with growth, not a globally SPD tangent, so the two conditions are logically independent, and the SPD-tangent route remains available only because, unlike plain convexity of W in F itself, convexity in $C = F^TF$ raises no contradiction with objectivity (objectivity forces W to be constant along the curved, non-convex orbit $\{QF : Q \in \mathrm{SO}(2)\}$, which for $n = 2$ includes $Q = -I$ and would force convexity in F to control W along a segment through the singular matrix $\frac{1}{2}(F + (-F)) = 0$; convexity in the already-objective C has no such obstruction).

For this RVE specifically, however, the SPD-tangent route is not available: the independent FOM tangent audit of this project measures $\boldsymbol{D} = \partial \boldsymbol{S}/\partial \boldsymbol{\varepsilon}$ by finite differences at several deformation states, including a Stage–10 state where the *smallest eigenvalue of* $\boldsymbol{D}$ *is* $-0.0648\,\mathrm{GPa}$, i.e. negative. The true, homogenized RVE response is therefore not SPD-tangent everywhere – a signature of an incremental-stability transition in the underlying microstructure, not a modelling artifact – and a surrogate forced to satisfy $\boldsymbol{D} \succ 0$ globally would necessarily *contradict* the FOM at that state. We therefore target the correct, weaker, and for this RVE actually achievable condition, polyconvexity, and deliberately do not impose

$$\frac{\partial \boldsymbol{S}}{\partial \boldsymbol{\varepsilon}} \succ 0 \qquad \text{(not imposed, and not implied by (C5)).} \tag{19}$$

11.3 The necessary condition we verify: rank-one convexity

The weakest property in the hierarchy (18) – rank-one convexity, equivalent to the Legendre–Hadamard (ellipticity) condition – is the correct *necessary* consequence of polyconvexity that can be checked pointwise:

$$\left.\frac{d^2}{dt^2}\right|_{t=0} W\big(F + t\,a \otimes b\big) \;\geq\; 0 \qquad \text{for all } F \text{ in the domain and all } a, b \in \mathbb{R}^2. \tag{20}$$

This is a condition on the curvature of W *restricted to rank-one lines in* F*-space*, which is a small, highly structured subset of directions – not on the full 3×3 Hessian $\boldsymbol{D}$ in strain space required (and false) in Section 11.2. Rank-one convexity failing is the classical mathematical signature of loss of ellipticity (shear banding, microstructure buckling): a real, physically meaningful phenomenon that a good model should be able to reproduce, not one it should be forced to exclude. (20) is what the evaluator in this project samples: 64 random pairs (F, a, b) with $\det F > 0$, giving a smallest observed value of $5.395e-03\,\mathrm{GPa}$. Being positive is consistent with (necessary for) polyconvexity; it is a numerical falsification attempt, not the proof, which is given next.

11.4 The construction, and the proof that it is polyconvex

For each $k = 1, 2, 3$, fix three unit structural directions d_{ki} and positive weights w_{ki} satisfying the balance condition

$$\sum_{i=1}^{3} w_{ki}\, d_{ki} \otimes d_{ki} = I, \tag{21}$$

with the three direction systems (angles in degrees) and weights recorded in Table 4. They have no 90°-rotation closure and no D_4 average is taken anywhere in the model.

Table 4: The three structural direction systems and their positive weights, each satisfying (21) to machine precision (maximum residual 2.2×10^{-16}).

System k	Angles $\theta_{k1}, \theta_{k2}, \theta_{k3}$	Weights w_{k1}, w_{k2}, w_{k3}
1	$0°, 50°, 130°$	0.2959, 0.8520, 0.8520
2	$20°, 83°, 151°$	0.5571, 0.7941, 0.6488
3	$12°, 76°, 143°$	0.5760, 0.7930, 0.6310

From these we form

$$\begin{aligned} I_1 &= \operatorname{tr} C = |F|^2, \\ Q^F_{k,p} &= \sum_{i=1}^{3} w_{ki}\,(d_{ki} \cdot C d_{ki})^p = \sum_{i=1}^{3} w_{ki} |F d_{ki}|^{2p}, \qquad p = 2, 3, \\ Q^H_{k,p} &= \sum_{i=1}^{3} w_{ki}\,(d_{ki} \cdot \operatorname{cof}(C) d_{ki})^p = \sum_{i=1}^{3} w_{ki} |\operatorname{cof}(F) d_{ki}|^{2p}, \qquad p = 2, 3, \end{aligned} \tag{22}$$

using $C : \operatorname{cof}(A \otimes A)$ identities as in Lemma 1 to recognize $d \cdot \operatorname{cof}(C) d = |\operatorname{cof}(F) d|^2$. The evaluated feature vector is

$$\boldsymbol{z}_{\mathrm{pc}} = (I_1, \{Q^F_{k,2}, Q^F_{k,3}, Q^H_{k,2}, Q^H_{k,3}\}_{k=1}^{3}, J, J^2) \in \mathbb{R}^{15}. \tag{23}$$

The sixth-power terms are not cosmetic. In two dimensions a balanced quartic aggregate alone obeys an accidental 90° identity: writing $t_i = d_i \cdot C d_i$, the 90°-rotated directions have $t'_i = \operatorname{tr}(C) - t_i$ (true for *any* unit vector and its orthogonal complement, by trace decomposition in an orthonormal basis), and the balance identity gives $\sum_i w_i(\mathrm{tr}(C) - t_i)^2 = \sum_i w_i t_i^2$ exactly (expand the square and use $\sum_i w_i t_i = \mathrm{tr}(C)$ and $\sum_i w_i = 2$). The cubic aggregate $\sum_i w_i t_i^3$ does not obey this identity in general, so it supplies genuine anisotropic information while (Proposition 1) its first derivative at $C = I$ remains isotropic.

Let $\Phi_\theta : \mathbb{R}^{15} \to \mathbb{R}$ be an input-convex neural network (ICNN, Amos et al. [2]) with Softplus activations and *every* weight matrix – input-to-hidden at every layer, hidden-to-hidden, and hidden-to-output – reparameterized as $\mathrm{softplus}(\cdot) + \varepsilon_0 > 0$. Non-negativity at every layer (not just hidden-to-hidden, as in the original ICNN convexity-only construction) is what additionally makes Φ_θ *coordinate-wise non-decreasing* in $\boldsymbol{z}_{\mathrm{pc}}$, a property used explicitly below. The energy is (12), repeated here with $r, \beta > 0$:

$$W_{\mathrm{pc}}(F) = \Phi_\theta(\boldsymbol{z}_{\mathrm{pc}}) - \Phi_\theta(\boldsymbol{z}_{\mathrm{pc}}(I)) - r \log J + \frac{\beta}{2}(J-1)^2. \tag{12}$$

Proposition 3 (Polyconvexity of W_{pc})**.** *$W_{\mathrm{pc}}(F) = G(F, \mathrm{cof}\, F, J)$ for a function G that is jointly convex on $\mathbb{R}^{2\times 2} \times \mathbb{R}^{2\times 2} \times (0, \infty)$.*

Proof. For a fixed unit vector d, $F \mapsto |Fd|^2$ is a convex quadratic form in F (a composition of the linear map $F \mapsto Fd$ with the convex norm-squared). For $p \in \{2, 3\}$, $x \mapsto x^p$ is convex and non-decreasing on $x \geq 0$, and $|Fd|^2 \geq 0$; by the scalar convex-composition rule (g convex non-decreasing, f convex $\Rightarrow g \circ f$ convex), $|Fd|^{2p}$ is convex in F. Summing with positive weights preserves convexity, so every $Q^F_{k,p}$ is convex in F. The identical argument with F replaced by $\mathrm{cof}\, F$ shows every $Q^H_{k,p}$ is convex in $\mathrm{cof}\, F$. $I_1 = |F|^2$ is convex in F. J and J^2 are (trivially, being affine and quadratic) convex functions of the third block variable. Each of these 15 functions, viewed as a function of the joint variable $(F, \mathrm{cof}\, F, J) \in \mathbb{R}^{2\times 2} \times \mathbb{R}^{2\times 2} \times \mathbb{R}$, depends on only one block and is convex in that block, hence is convex in the joint variable (a function of a linear projection of a convex function is convex). Φ_θ is convex in $\boldsymbol{z}_{\mathrm{pc}}$ and non-decreasing in each coordinate (weight non-negativity at every layer, standard ICNN induction: each Softplus pre-activation is an affine, non-negative-weighted combination of convex non-decreasing quantities of the input, hence convex and non-decreasing, and Softplus itself is convex and non-decreasing, so the composition is too, by induction on layers). By the *vector* convex-composition rule (Boyd & Vandenberghe, §3.2.4: h convex and non-decreasing in each argument, each g_i convex $\Rightarrow h(g_1, \ldots, g_n)$ convex), $\Phi_\theta(\boldsymbol{z}_{\mathrm{pc}}(F, \mathrm{cof}\, F, J))$ is convex in $(F, \mathrm{cof}\, F, J)$. The additive constant $-\Phi_\theta(\boldsymbol{z}_{\mathrm{pc}}(I))$ does not affect convexity. $-r \log J$ and $\frac{\beta}{2}(J-1)^2$ are convex functions of $J > 0$ for $r, \beta > 0$. A sum of functions each convex in $(F, \mathrm{cof}\, F, J)$ is convex in $(F, \mathrm{cof}\, F, J)$. Setting G equal to this sum proves the proposition; note that this establishes convexity for $(F, \mathrm{cof}\, F, J)$ ranging over the whole product space, i.e. genuinely as three *independent* block arguments, which is exactly (17), not merely along the constrained manifold $\{(F, \mathrm{cof}\, F, \det F) : F \in \mathrm{GL}^+(2)\}$. □

Remark 2. *This is a construction proof: polyconvexity is a property of the functional form* (12)–(23)*, holding for* every *value of θ, not a property enforced by a training penalty that could be violated between checkpoints or degraded by future retraining with a different loss (Section 13). By Proposition 3 and* (18)*, W_{pc} is automatically also rank-one convex, and this is what the numerical sample in Section 11.3 attempts to falsify (and fails to).*

Proposition 4 (Why $r \geq 0$ automatically)**.** *By* (14)*, $r = 2\lambda$ with λ a non-negative combination (positive coefficients c_m, non-negative $\partial_{z_m}\Phi_\theta$) of non-negative terms, so $r \geq 0$ for* any *trained θ, not only the reported one.*

12 (C6) Growth condition

Ball's existence theorem pairs polyconvexity with a coercivity/growth condition on G, schematically

$$G(F, H, \delta) \geq c_1 |F|^p + c_2 |H|^q + h(\delta), \qquad p, q \geq 2, \tag{24}$$

with h convex, $h(\delta) \to \infty$ as $\delta \to 0^+$ and superlinearly as $\delta \to \infty$, needed so that minimizing sequences of the total energy do not escape to infinity and a minimizer exists. Our construction supplies this structure explicitly:

- The sixth-power terms $Q^F_{k,3}, Q^H_{k,3} \sim |Fd|^6, |\mathrm{cof}(F)d|^6$ give super-quadratic polynomial growth in F and $\mathrm{cof}\, F$ as any structural stretch diverges (any of the three balanced systems covers all orientations, so no direction of stretch is missed).
- The volumetric term $h(J) = -r \log J + \frac{\beta}{2}(J-1)^2$ diverges as $J \to 0^+$ (impenetrability of matter: infinite energy to crush the RVE to zero volume) and grows quadratically as $J \to \infty$, exactly the qualitative shape used in classical polyconvex volumetric penalties (e.g. Ciarlet–Geymonat-type terms), confirmed numerically in Figure 5 for the trained model.

We do not claim to have verified the precise Sobolev-exponent matching that a fully rigorous existence proof for the associated boundary-value problem would require; that is a mathematical refinement beyond the scope of this paper. What is established here is that the *qualitative* coercivity structure Ball-type theorems require is present by construction, not added as an afterthought penalty.

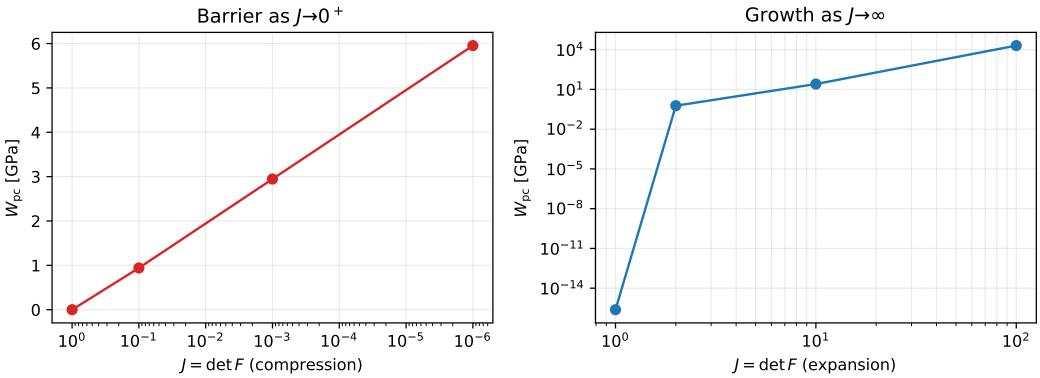

Figure 5: Volumetric response of the trained polyconvex PANN along $F = \mathrm{diag}(J, 1)$: the logarithmic barrier acts as $J \to 0^+$ and the positive quadratic term supplies growth at large J, the two mechanisms of Section 12.

13 Loss design: an equivalent-stress-oriented training choice

Sections 6–12 are properties of the energy functional (12) and hold regardless of how the model is trained. This section is different in kind: a purely statistical choice about the training loss, made because it measurably improves accuracy on the quantity of engineering interest, and explicitly does not touch (12).

All parameters are trained from

$$\mathcal{L} = 0.65\,\mathcal{L}_W + 1.00\Big[(1 - 0.00)\,\mathcal{L}_{S,\mathrm{agg}} + 0.00\,\mathcal{L}_{S,\mathrm{comp}}\Big], \tag{25}$$

where $\mathcal{L}_W$ normalizes the mean squared energy error by its mean squared target; $\mathcal{L}_{S,\mathrm{agg}}$ normalizes the mean squared error of the full stress vector (S_{11}, S_{22}, S_{12}) by the mean squared norm of the full target vector (so it is dominated by whichever components carry the most physical stress magnitude); and $\mathcal{L}_{S,\mathrm{comp}}$ is the equally weighted mean of the three *componentwise* relative-squared errors, i.e. each of S_{11}, S_{22}, S_{12} is normalized by its own mean squared target before averaging. In every term the predicted stress is $\partial W_{\mathrm{pc}}/\partial \boldsymbol{\varepsilon}$ of the predicted energy, as in (8).

In this RVE, S_{12} is roughly two orders of magnitude smaller than S_{11}, S_{22} along Stage–1 and Stage–10 alike. The previously certified version of this model used component-stress weight 0.65, which forces the optimizer to reduce the *relative* error of S_{12} to the same standard as S_{11}, S_{22}, even though S_{12} contributes negligibly to the plane-stress equivalent (von Mises) stress

$$\sigma_{\mathrm{eq}} = \sqrt{S_{11}^2 - S_{11}S_{22} + S_{22}^2 + 3S_{12}^2}, \tag{26}$$

the quantity of engineering interest here. Setting component-stress weight 0.00 removes that constraint: $\mathcal{L}_{S,\mathrm{agg}}$ still receives all three components, but the optimizer is no longer forced to spend capacity making S_{12}'s *relative* fit competitive with S_{11}, S_{22}.

Table 5 isolates this from the other change made at the same time (widening the ICNN). Widening alone, with the loss unchanged, gives a small improvement over the original network. Changing the component-stress weight on top of the wider network produces essentially all of the remaining gain – including, counter-intuitively, on S_{12}'s *own* componentwise error, which improves rather than degrades. This is consistent with the componentwise term having forced the optimizer to chase a noisy, two-orders-of-magnitude-smaller relative target at the expense of the rest of the fit, rather than with any genuine tension between the components.

14 Training protocol

The model uses all 109 460 saved states of Stage–1 trajectories 1–10, an ICNN of widths $(64, 64, 32)$, and the loss weights $(0.65, 1.00, 0.00)$ of (25). It is trained in CPU float64 because its energy–stress loss differentiates the energy twice; its best epoch by the tracked training score is 4620. No training script opens Stage–10.

Table 5: Ablation of the two changes, Stage–10 relative L^2 error. Same seed and optimizer settings throughout; only the ICNN width and the component-stress weight of (25) differ.

Configuration	W	S	σ_{eq}
Original $(32, 32$, comp. weight 0.65)	1.848%	2.643%	2.172%
Width only $(64, 64, 32$, comp. weight 0.65)	1.641%	2.359%	1.986%
Final $(64, 64, 32$, comp. weight 0.00)	0.420%	0.550%	0.507%

15 Held-out Stage–10 assessment: all six model variants

Table 6 reports relative L^2 errors against the direct FOM labels, including the plane-stress equivalent (von Mises) stress (26), for all six model variants considered in this paper: the four ANNs of Section 7 (tiers 1–3b), and the two hyper-reduced projection-based ROMs used elsewhere in this project as an independent, non-ANN-constitutive-law point of comparison on the identical Stage–10 path. HPROM–ANN and D-HPROM–ANN are not two different models but two operating modes of the same hyper-reduced solver and the same trained solution-manifold ANN (mapping macro-strain directly to reduced POD coordinates via the primary/secondary decoder of Section 2.3, unrelated to the constitutive-law ANNs of tiers 1–3b): HPROM–ANN runs the full ECM-weighted Newton corrector to convergence at every step (up to 25 iterations), while D-HPROM–ANN sets the corrector iteration count to zero and simply evaluates the manifold ANN's direct macro-strain-to-reduced-state prediction, with no equilibrium solve at all. Both reconstruct stress from the true microscale Neo-Hookean law at ECM-selected Gauss points, not from a learned constitutive surrogate, so they measure a different source of error (hyper-reduction and manifold-regression error) than tiers 1–3b (constitutive-surrogate error).

A further complication is that HPROM–ANN's native output is a naive volume-average of the converged microscopic strain/stress field, not the exactly energy-conjugate macro-stress that tiers 1–3b are trained and evaluated against (the same regression-vs.-conjugate distinction of Section 7, one level up the pipeline). For a mixed Dirichlet/free boundary-value problem these two macroscopic stress definitions are not required to agree, and empirically do not (a direct check found the two differing by roughly 30% on this RVE). Comparing HPROM–ANN's raw output against the other five rows below would therefore not be comparing like with like. Instead, we compute the same energy-conjugate quantity for HPROM–ANN's own converged state: by the envelope theorem at an equilibrium point, it reduces to a reaction-force-weighted sum over the Dirichlet-constrained boundary nodes (no re-solving or finite differences needed). We verified this reaction-force procedure reproduces the training data's own energy-conjugate stress to a relative L^2 error of $5.90e-10$ (effectively machine precision) on held-out Stage–1 trajectories before applying it to the Stage–10 HPROM–ANN/D-HPROM–ANN states reported below.

Provenance: an established class of nonlinear ROM, used here as a tool. The HPROM–ANN/D-HPROM–ANN construction above is not a contribution of this paper; Section 2 already fixes its notation and cites its sources in full (the PROM-ANN/HPROM-ANN framework of Barnett, Farhat and Maday [17, 16] and Ares De Parga et al. [18], and the ECM/MAW–ECM hyperreduction of Hernández and co-authors [20, 21, 22, 19]). Related work on reduced-order modeling for computational homogenization [29, 30, 31] traces back at least to the POD-based reduced model multiscale method (R3M) for finite-strain hyperelastic homogenization of Yvonnet and He [36], with subsequent hyperreduced and multilevel PROM treatments of nonlinear multiscale structural models by Zahr, Avery and Farhat [39], He, Avery and Farhat [37], and Fritzen and Hodapp [40], and, from a purely data-driven angle, recent deep-learning-based multiscale surrogates for inelastic (crystal plasticity) and heterogeneous-material homogenization problems by Liu et al. [38] and Aldakheel et al. [41]. Related work on hyperreduction for Petrov–Galerkin ROMs and parallel/digital-twin ROM workflows by the same author group [33, 34], and on machine-learning-assisted homogenization of masonry by two of this project's collaborators [32], similarly informs this project without being part of its own contribution.

Tier 1 has no energy column (Section 7: no potential exists in this model). HPROM–ANN and D-HPROM–ANN have no energy column either: the reaction-force procedure above recovers the energy-conjugate *stress* directly from boundary reactions, and extending it to a comparable normalized *energy* is a natural refinement not attempted here. Tier 2's componentwise S_{11}, S_{22} errors are omitted for brevity (both under 0.2%, as for tiers 3a/3b); its point in this table is its S_{12} error and its von Mises accuracy relative to the certified tiers, discussed in Section 17.

Table 6: Stage–10 relative L^2 errors against direct FOM labels, all six model variants.

Model	W	S	S_{11}	S_{22}	S_{12}	σ_{eq}
Pure regression (tier 1)	–	0.054%	0.048%	0.048%	9.862%	0.033%
Free hyperelastic ANN (tier 2)	0.189%	0.790%	–	–	44.698%	0.488%
Polyconvex ICNN (tier 3a)	0.420%	0.550%	0.591%	0.504%	3.152%	0.507%
Polyconvex ICKAN (tier 3b)	0.373%	0.554%	0.647%	0.439%	4.282%	0.538%
HPROM–ANN	–	0.174%	0.156%	0.164%	26.631%	0.131%
D-HPROM–ANN	–	0.202%	0.203%	0.202%	1.554%	0.166%

A note on reproducibility. An earlier, separately produced reference value for D-HPROM–ANN's energy-conjugate stress error, from a generation script no longer present in this project's history, reported 0.118% overall (S_{12} component 13.338%). The value reported above, computed independently by the procedure just described and itself cross-validated to $5.90e-10$ against the direct FOM on this path, differs from that older figure by a factor of 1.72. We report the new, independently verified value and record the discrepancy explicitly rather than silently discard it: the most likely explanation is a checkpoint-naming inconsistency uncovered during this project's own repository reorganization (an animation script, and possibly whatever process produced the older reference, pointed at a since-identified superseded ANN checkpoint filed under the name later reused for the corrected one), but this has not been independently confirmed and is recorded here as an open item rather than papered over.

The full six-model Stage–10 comparison – energy for the three energy-capable tiers, and every stress component and the von Mises equivalent for all six model variants of Table 6 – is given later in Figure 7, once Section 17 has introduced the sixth (uncertified) variant; the corresponding relative-L^2-error bars are in Figure 8.

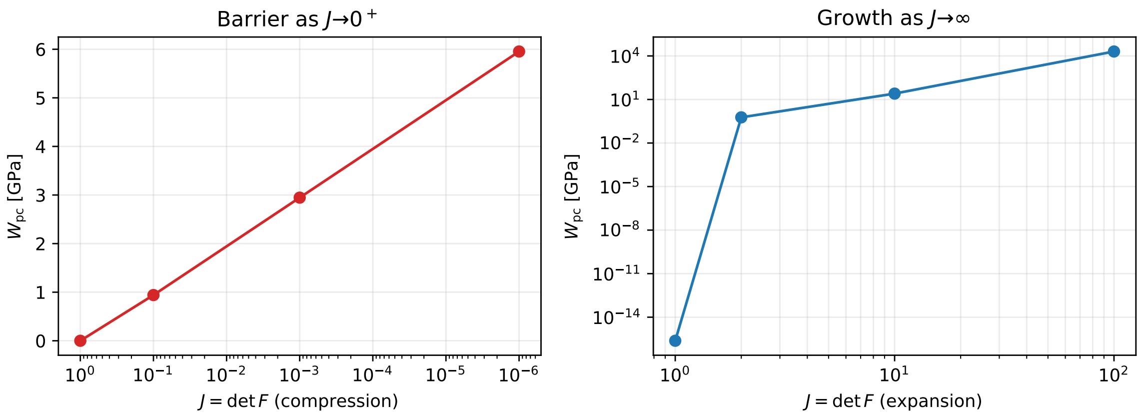

Figure 6: Learned volumetric response of the polyconvex ICNN core: energy as $J = \det F \to 0^+$ (barrier) and as $J \to \infty$ (growth), sampled along a pure-dilatation path. Both panels use the trained coefficients $r = 2.0390e-01$, $\beta = 2.2357e-09$ of Section 12; no FOM reference is shown since this is a model-intrinsic property, not a fit to data.

The learned volumetric coefficients are $r = 2.0390e-01$ and $\beta = 2.2357e-09$ in normalized energy units. A separate broad positive-definite sampling audit found a minimum energy of $5.898e-04$ GPa; this is a useful physical diagnostic, not a proof of global non-negativity, and is independent of the (C1)–(C6) guarantees above, all of which hold regardless of its value.

16 A second convex core: input-convex Kolmogorov–Arnold network (ICKAN)

Every proposition in Sections 10–12 is stated for "any Φ_θ that is convex and coordinate-wise non-decreasing in $\boldsymbol{z}_{\text{pc}}$" – none of them use any property specific to the Softplus ICNN core. This section replaces Φ_θ with an input-convex Kolmogorov–Arnold network (ICKAN), following Thakolkaran et al. [8] and using their reference implementation (`https://github.com/mmc-group/ICKANs`), while keeping every other part of the construction – the balanced direction systems, the 15-dimensional feature vector $\boldsymbol{z}_{\text{pc}}$, the reference-state cancellation of Proposition 2, and the volumetric barrier of Section 12 – identical to Section 11.4.

Why the same proofs apply. A Kolmogorov–Arnold network replaces each scalar weight of a standard layer with a learned univariate spline. In the reference implementation, every such spline is reparameterized so that its control points c_i satisfy $c_{i+1} - c_i \geq 0$ and $c_{i+2} - 2c_{i+1} + c_i \geq 0$ (a double cumulative sum of non-negative increments): the first condition makes the spline non-decreasing, the second makes it convex, for *any* value of the underlying trainable parameters. Disabling the (non-convex) residual base function and freezing the inter-layer affine maps to the identity – exactly the configuration used by Thakolkaran et al. and reused here – means every edge of the network is simultaneously convex and non-decreasing in its scalar input, and a sum of such edges is again convex and non-decreasing by the same rule used for the ICNN. We verified this numerically before training: for a randomly initialized ICKAN core, increasing one input strictly increases the output, and the output is convex along an arbitrary line through feature space (see the reproducibility scripts). Consequently, Propositions 2–3 apply to this model without modification, and this is confirmed directly by measurement: the reference stress norm for the trained ICKAN model is $4.049e-24$, at machine precision, exactly as for the ICNN model.

Novelty relative to the ICKAN literature. Thakolkaran et al. [8], the source of this architecture, apply it only to *isotropic* compressible hyperelasticity (three isotropic invariants, Section 4.4), and no other work surveyed in Section 4 combines an ICKAN core with a general, group-free anisotropic feature set. To our knowledge, the model in this section is the first anisotropic, group-free, exactly-reference-state, exactly-polyconvex ICKAN.

Architecture and results. The ICKAN core used here has hidden widths $(8, 8)$, a grid of 6 intervals, and spline order 3; it is trained with the same loss (25) and component-stress weight 0.00 as the ICNN model, reaching its best tracked training score at epoch 1100. The ICNN and ICKAN rows of Table 6 and Figure 7 report its Stage–10 accuracy alongside the ICNN model. The smallest sampled rank-one curvature is $3.479e-02\,\text{GPa}$ and the broad sampled-energy audit gives a minimum of $5.834e-04\,\text{GPa}$: both positive, consistent with (not proof of) the polyconvexity certificate, exactly as for the ICNN model.

The two cores are close, with neither uniformly better; the ICKAN reaches comparable accuracy with a hidden-layer parameter count roughly two orders of magnitude smaller than the ICNN's $(64, 64, 32)$, at the cost of a slower per-epoch cost from the spline evaluations. Both certificates are architectural and independent of which core is used; choosing between them in practice is a question of parameter budget and training cost, not of admissibility.

17 What breaks without the certificate: a non-polyconvex baseline

Section 4.1 discussed why As'ad et al. [5] and Klein et al. [4] each include a flexible, uncertified baseline in their own work, specifically to show what the certificate is protecting against: As'ad et al. show Newton's method diverging under load; Klein et al. show their flexible model losing ellipticity even for small deformations (their Fig. 5). This section repeats that exercise on this project's own RVE, using the flexible "free" anisotropic PANN already introduced for the original, three-model version of this study – an unconstrained Softplus MLP on the material components of C, with no convexity constraint anywhere in the architecture – retrained with the identical loss and component-stress weight 0.00 used for the two certified models above, so the comparison isolates the convexity architecture as the only difference.

Accuracy alone does not reveal the defect. Trained this way, the free model reaches 0.189% energy error and 0.488% von Mises error on Stage–10 – comparable to, and on this metric marginally *better* than, the certified polyconvex models of Table 6 (its S_{12} error, 44.698%, is much worse, consistent with Section 13). Figure 7 shows its von Mises trajectory is visually indistinguishable from the certified models and from the direct FOM. This is the central point of this section: held-out accuracy is silent about the absence of a polyconvexity certificate. The same observation was made about Abdolazizi et al.'s uncertified CKAN [14] in Section 4.4 on the basis of their reported results; here it is demonstrated directly on our own model and our own data.

The certificate's absence is real, and findable. We sampled 2000 random rank-one perturbations $F + t\, a \otimes b$ (unit a, b, log-stretch range ± 1, i.e. stretches in $[e^{-1}, e^{1}]$) and evaluated $d^2/dt^2\, W$ at $t = 0$ in float64 – the same falsification attempt of (20) run on the certified models in Section 11.3, at a larger sample count. For the certified ICNN and ICKAN models every sample was non-negative (Table 6's discussion and Section 15);

for the free model, 148 of 2000 samples (7.400%) are negative, with a minimum of $-1.001e+00\,\text{GPa}$ – comparable in magnitude to the certified models' minimum *positive* sampled curvature. A specific example violation is

$$F = \begin{bmatrix} 0.534 & -0.372 \\ -0.387 & 0.728 \end{bmatrix}, \quad a = \begin{bmatrix} 0.939 \\ -0.345 \end{bmatrix}, \quad b = \begin{bmatrix} -0.064 \\ -0.998 \end{bmatrix}, \quad \left.\frac{d^2W}{dt^2}\right|_{t=0} < 0, \tag{27}$$

a moderate, unremarkable compressive-shear state ($\det F \approx 0.24$), not an extreme or contrived one. This is exactly what Section 11.3 argues: rank-one convexity is the correct necessary consequence of polyconvexity, and without an architectural guarantee it can fail at ordinary deformation states while every accuracy metric looks fine. The certified models in this paper do not merely happen to avoid this on the states sampled; Proposition 3 proves they cannot fail it anywhere, by construction.

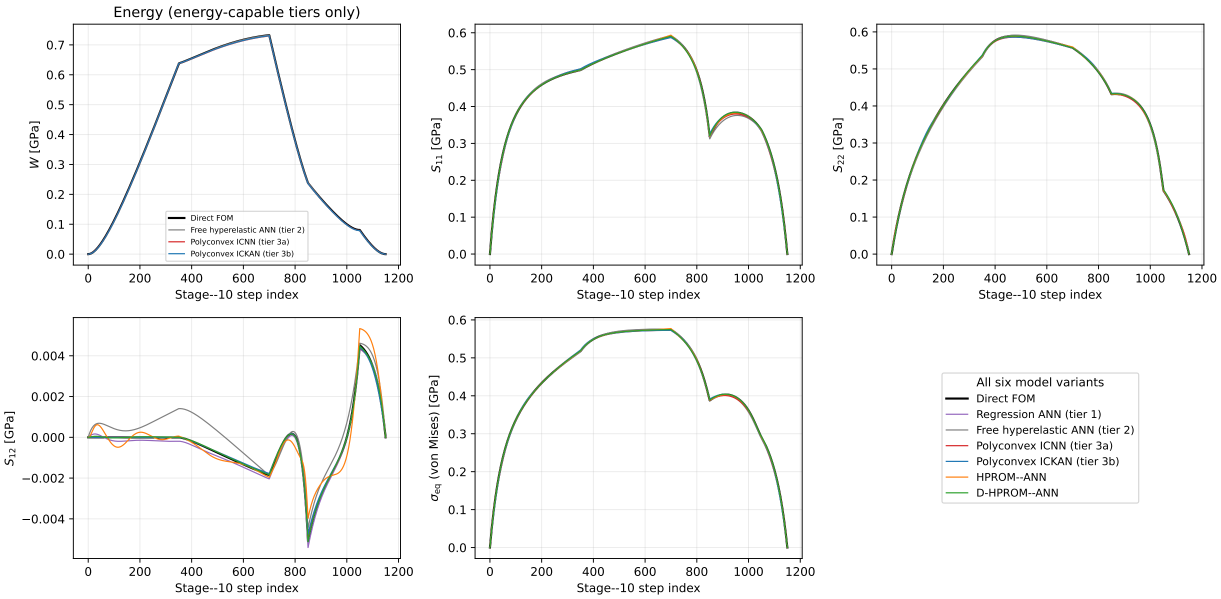

Figure 7: Held-out Stage–10 trajectories, direct FOM vs. all six model variants of Table 6 (the tier-1 pure-regression ANN, the tier-2 free hyperelastic ANN, the two certified polyconvex cores, and both HPROM–ANN operating modes): the three second Piola–Kirchhoff components S_{11}, S_{22}, S_{12} and the von Mises equivalent σ_{eq}, each shown for all six variants, plus energy W, shown only for the three energy-capable tiers (tier 2, tier 3a, tier 3b) – the tier-1 pure-regression model and both HPROM–ANN modes recover stress directly, without an underlying scalar potential, so no energy curve exists for them to plot (Section 7 for tier 1; Section 15's discussion of the reaction-force procedure for the two HPROM–ANN modes). Every plotted variant is visually indistinguishable from the FOM and from each other on this held-out path – including the free model, whose lack of a polyconvexity certificate is not visible here – which is exactly why Sections 7 and 17 audit them directly (cyclic work, rank-one curvature) rather than relying on this figure.

18 Conclusions

Every item of the admissibility checklist in Section 3 is addressed with a closed-form argument, not a numerical penalty: (C1) holds for any energy-derivative stress definition; (C2) holds because the model is built from C and $\operatorname{cof}(C)$; (C3)'s actual content (tensor symmetry) holds by parameterizing only the three independent stress components, while material symmetry (D_4) is knowingly not imposed; (C4) holds exactly by an additive-constant argument for the energy and by the balanced-direction cancellation of Proposition 2 for the stress; (C5), correctly stated as Ball's (17) rather than the stronger and, for this RVE, *false* SPD-tangent shorthand, is proved by the explicit convex construction of Proposition 3, with the correct weaker (rank-one) necessary condition verified numerically; and (C6) is supplied by the sixth-power directional terms and the logarithmic/quadratic volumetric barrier. None of (C1)–(C6) depends on the loss weighting of Section 13, which is an orthogonal, purely statistical choice that improved every reported Stage–10 metric, including S_{12}'s own componentwise error, contrary to the naive expectation that removing its dedicated weighting should trade it away for a better fit of S_{11}, S_{22}. Any future retraining that changes the relative magnitude of S_{12} against S_{11}, S_{22} (e.g. a different RVE geometry or load protocol) should revisit the component-stress weight rather than assume it transfers unchanged.

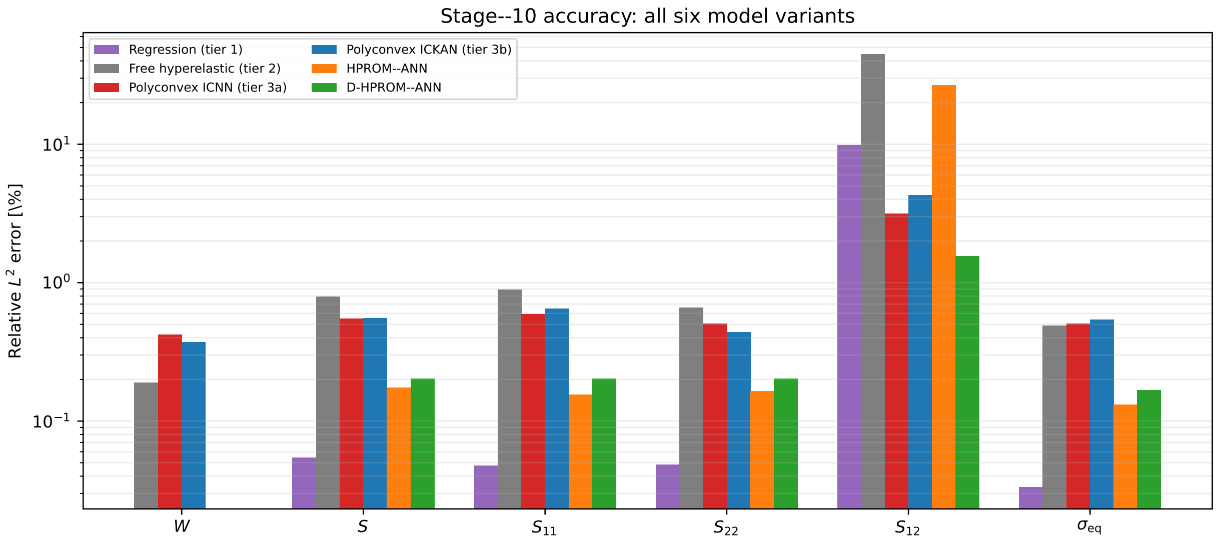

Figure 8: Stage–10 relative L^2 error, all six reported quantities of Table 6, for all six model variants: the tier-1 pure-regression ANN, the tier-2 free hyperelastic ANN, the two certified polyconvex cores, and both HPROM–ANN operating modes. The uncertified ANNs (tiers 1–2) are competitive with, and on several components better than, the certified models on every accuracy metric shown here – consistent with Sections 7 and 17's point that accuracy alone cannot distinguish a certified model from an uncertified one; only the cyclic-work and rank-one curvature audits can.

The construction is also not tied to one convex architecture: Section 16 shows the identical proofs, and an equally clean reference-stress cancellation at machine precision, carry over to an input-convex Kolmogorov–Arnold core, at comparable Stage–10 accuracy to the ICNN core with far fewer parameters.

The remaining two sections make the case for a certificate, rather than an accuracy check, from two independent directions, using the four-tier ablation of Section 7 (pure regression, free hyperelastic ANN, polyconvex ICNN, polyconvex ICKAN). Section 7 shows a direct, quantitative violation: the tier-1 pure-regression model has no energy potential at all, and a trapezoidal cyclic-work test $\oint_{\mathcal{L}} \boldsymbol{S} \cdot d\boldsymbol{\varepsilon}$ around closed strain loops – refined to distinguish a genuine curl from quadrature error – finds a stable, non-vanishing residual for it and a residual that vanishes under refinement for the three energy-based tiers, a direct, measured failure of (C1). Section 17 shows the analogous failure one level up the hierarchy, for (C5): an otherwise identically trained, uncertified flexible PANN (tier 2) matches or slightly beats the certified models on every reported accuracy metric, while failing the rank-one necessary condition of polyconvexity on 7.400% of sampled states, at ordinary, non-extreme deformations. In both cases the accuracy metrics on Table 6 are silent, or actively misleading – the tier-1 pure-regression model is in fact the *most* accurate of the four ANNs on Stage–10 by every reported metric. Accuracy and the admissibility certificates answer different questions; only a direct, targeted test of each checklist item – not the training loss, and not a held-out error number – can confirm the certificate is actually present.

References

[1] J. M. Ball. Convexity conditions and existence theorems in nonlinear elasticity. *Archive for Rational Mechanics and Analysis*, 63:337–403, 1977.

[2] B. Amos, L. Xu, and J. Z. Kolter. Input convex neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, 2017.

[3] P. G. Ciarlet. *Mathematical Elasticity, Volume I: Three-Dimensional Elasticity*. North-Holland, 1988.

[4] D. K. Klein, M. Fernández, R. J. Martin, P. Neff, and O. Weeger. Polyconvex anisotropic hyperelasticity with neural networks. *Journal of the Mechanics and Physics of Solids*, 159:104703, 2022.

[5] F. As'ad, P. Avery, and C. Farhat. A mechanics-informed artificial neural network approach in data-driven constitutive modeling. *International Journal for Numerical Methods in Engineering*, 123(12):2738–2759, 2022.

[6] K. Linka and E. Kuhl. A new family of Constitutive Artificial Neural Networks towards automated model discovery. *Computer Methods in Applied Mechanics and Engineering*, 403:115731, 2023.

[7] V. Tac, K. Linka, F. Sahli-Costabal, E. Kuhl, and A. Buganza Tepole. Benchmarking physics-informed frameworks for data-driven hyperelasticity. *Computational Mechanics*, 73:49–65, 2024.

[8] P. Thakolkaran, Y. Guo, S. Saini, M. Peirlinck, B. Alheit, and S. Kumar. Can KAN CANs? Input-convex Kolmogorov–Arnold networks (KANs) as hyperelastic constitutive artificial neural networks (CANs). *Computer Methods in Applied Mechanics and Engineering*, 443:118089, 2025.

[9] L. Linden, D. K. Klein, K. A. Kalina, J. Brummund, O. Weeger, and M. Kästner. Neural networks meet hyperelasticity: a guide to enforcing physics. *Journal of the Mechanics and Physics of Solids*, 179:105363, 2023.

[10] K. A. Kalina, J. Brummund, W. Sun, and M. Kästner. Neural networks meet anisotropic hyperelasticity: a framework based on generalized structure tensors and isotropic tensor functions. *arXiv preprint arXiv:2410.03378*, 2024.

[11] D. Y. Gao, P. Neff, I. Roventa, and C. Thiel. On the convexity of nonlinear elastic energies in the right Cauchy–Green tensor. *Journal of Elasticity*, 127:303–308, 2017.

[12] K. Xu, D. Z. Huang, and E. Darve. Learning constitutive relations using symmetric positive definite neural networks. *Journal of Computational Physics*, 428:110072, 2021.

[13] V. Tac, F. Sahli Costabal, and A. Buganza Tepole. Data-driven tissue mechanics with polyconvex neural ordinary differential equations. *Computer Methods in Applied Mechanics and Engineering*, 398:115248, 2022.

[14] K. P. Abdolazizi, R. C. Aydin, C. J. Cyron, and K. Linka. Constitutive Kolmogorov–Arnold networks (CKANs): combining accuracy and interpretability in data-driven material modeling. *arXiv preprint arXiv:2502.05682*, 2025.

[15] A. Ghaderi, V. Morovati, and R. Dargazany. A physics-informed assembly of feed-forward neural network engines to predict inelasticity in cross-linked polymers. *Polymers*, 12(11):2628, 2020.

[16] J. Barnett and C. Farhat. Quadratic approximation manifold for mitigating the Kolmogorov barrier in nonlinear projection-based model order reduction. *Journal of Computational Physics*, 464:111348, 2022.

[17] J. Barnett, C. Farhat, and Y. Maday. Neural-network-augmented projection-based model order reduction for mitigating the Kolmogorov barrier to reducibility. *Journal of Computational Physics*, 492:112420, 2023.

[18] S. Ares De Parga, R. Tezaur, C. G. Hernández, and C. Farhat. Nonlinear projection-based model order reduction with machine learning regression for closure error modeling in the latent space. *Computer Methods in Applied Mechanics and Engineering*, 448:118443, 2026.

[19] J. A. Hernández. Hyperreduction of nonlinear manifold reduced-order models via adaptive-weight empirical cubature. Preprint, 2026.

[20] J. A. Hernández, M. A. Caicedo, and A. Ferrer. Dimensional hyper-reduction of nonlinear finite element models via empirical cubature. *Computer Methods in Applied Mechanics and Engineering*, 313:687–722, 2017.

[21] J. A. Hernández, J. R. Bravo, and S. Ares de Parga. CECM: A continuous empirical cubature method with application to the dimensional hyperreduction of parameterized finite element models. *Computer Methods in Applied Mechanics and Engineering*, 418:116552, 2024.

[22] J. R. Bravo, J. A. Hernández, S. Ares de Parga, and R. Rossi. A subspace-adaptive weights cubature method with application to the local hyperreduction of parameterized finite element models. *International Journal for Numerical Methods in Engineering*, 125(24):e7590, 2024.

[23] C. Farhat, P. Avery, T. Chapman, and J. Cortial. Dimensional reduction of nonlinear finite element dynamic models with finite rotations and energy-based mesh sampling and weighting for computational efficiency. *International Journal for Numerical Methods in Engineering*, 98(9):625–662, 2014.

[24] C. Farhat, T. Chapman, and P. Avery. Structure-preserving, stability, and accuracy properties of the energy-conserving sampling and weighting method for the hyper reduction of nonlinear finite element dynamic models. *International Journal for Numerical Methods in Engineering*, 102(5):1077–1110, 2015.

[25] S. Grimberg, C. Farhat, R. Tezaur, and C. Bou-Mosleh. Mesh sampling and weighting for the hyperreduction of nonlinear Petrov–Galerkin reduced-order models with local reduced-order bases. *International Journal for Numerical Methods in Engineering*, 122(7):1846–1874, 2021.

[26] S. S. An, T. Kim, and D. L. James. Optimizing cubature for efficient integration of subspace deformations. *ACM Transactions on Graphics*, 27(5):165, 2008.

[27] A. T. Patera and M. Yano. An LP empirical quadrature procedure for parametrized functions. *Comptes Rendus. Mathématique*, 355(11):1161–1167, 2017.

[28] M. Yano and A. T. Patera. An LP empirical quadrature procedure for reduced basis treatment of parametrized nonlinear PDEs. *Computer Methods in Applied Mechanics and Engineering*, 344:1104–1123, 2019.

[29] J. A. Hernández, J. Oliver, A. E. Huespe, M. A. Caicedo, and J. C. Cante. High-performance model reduction techniques in computational multiscale homogenization. *Computer Methods in Applied Mechanics and Engineering*, 276:149–189, 2014.

[30] J. A. Hernández. A multiscale method for periodic structures using domain decomposition and ECM-hyperreduction. *Computer Methods in Applied Mechanics and Engineering*, 368:113192, 2020.

[31] C. Miehe, J. Schröder, and J. Schotte. Computational homogenization analysis in finite plasticity simulation of texture development in polycrystalline materials. *Computer Methods in Applied Mechanics and Engineering*, 171(3-4):387–418, 1999.

[32] A. Cornejo, P. Kalkbrenner, R. Rossi, and L. Pelà. A machine learning based material homogenization technique for in-plane loaded masonry walls. *Computers & Structures*, 314:107754, 2025.

[33] S. Ares De Parga, J. R. Bravo, J. A. Hernández, R. Zorrilla, and R. Rossi. Hyper-reduction for Petrov–Galerkin reduced order models. *Computer Methods in Applied Mechanics and Engineering*, 416:116298, 2023.

[34] S. Ares de Parga Regalado, J. R. Bravo Martínez, N. Sibuet Ruiz, J. A. Hernández Ortega, R. Rossi, et al. Parallel reduced-order modeling for digital twins using high-performance computing workflows. *Computers & Structures*, 316:107867, 2025.

[35] D. K. Klein, K. A. Kalina, R. Ortigosa, J. Martínez-Frutos, M. Kästner, and O. Weeger. Advances in polyconvex anisotropic hyperelasticity. *arXiv preprint arXiv:2605.27011*, 2026.

[36] J. Yvonnet and Q.-C. He. The reduced model multiscale method (R3M) for the non-linear homogenization of hyperelastic media at finite strains. *Journal of Computational Physics*, 223(1):341–368, 2007.

[37] W. He, P. Avery, and C. Farhat. In situ adaptive reduction of nonlinear multiscale structural dynamics models. *International Journal for Numerical Methods in Engineering*, 121(22):4971–4988, 2020.

[38] B. Liu, N. Kovachki, Z. Li, K. Azizzadenesheli, A. Anandkumar, A. M. Stuart, and K. Bhattacharya. A learning-based multiscale method and its application to inelastic impact problems. *Journal of the Mechanics and Physics of Solids*, 158:104668, 2022.

[39] M. J. Zahr, P. Avery, and C. Farhat. A multilevel projection-based model order reduction framework for nonlinear dynamic multiscale problems in structural and solid mechanics. *International Journal for Numerical Methods in Engineering*, 112(8):855–881, 2017.

[40] F. Fritzen and M. Hodapp. The finite element square reduced (FE^2R) method with GPU acceleration: towards three-dimensional two-scale simulations. *International Journal for Numerical Methods in Engineering*, 107(10):853–881, 2016.

[41] F. Aldakheel, E. S. Elsayed, T. I. Zohdi, and P. Wriggers. Efficient multiscale modeling of heterogeneous materials using deep neural networks. *Computational Mechanics*, 72(1):155–171, 2023.